

HELIX CLUSTER OS — IMPLEMENTATION PLAN

10,000+ Granular Tasks | 50-Week Roadmap | v1.0

HOW TO READ THIS DOCUMENT

Task Format

[PHASE].[SUB-PHASE].[TASK_NUMBER] | [TITLE] | [HOURS]h | [PRIORITY] | [SKILL]
Acceptance: [Specific, measurable acceptance criteria]
Dependencies: [List of task IDs that must complete first]
Risks: [Identified risks and mitigations]
Deliverable: [Concrete output artifact]

Priority Levels

- **P0:** Critical path — project cannot proceed without this
- **P1:** High priority — essential for MVP
- **P2:** Medium priority — important but can be deferred
- **P3:** Low priority — nice to have, post-MVP

Skill Tags

- **G0:** Go language development
- **ZIG:** Zig language development
- **C:** C/C++ development (including CUDA)
- **BASH:** Shell scripting
- **DB:** Database design and operations
- **NET:** Networking
- **SEC:** Security
- **ML:** Machine learning / LLM integration
- **QA:** Quality assurance and testing
- **OPS:** Operations and DevOps
- **DOC:** Documentation

PHASE 0: FOUNDATION (Weeks 1-4)

0.1 PROJECT BOOTSTRAP (Week 1)

0.1.1 Repository Structure

0.1.1.001 | Create monorepo root with directory structure | 2h | P0 | BASH
Acceptance: All directories exist: /cmd, /pkg, /internal, /api, /web, /docs, /scripts, /deploy, /test
Dependencies: None
Risks: None
Deliverable: Directory tree committed to main branch

0.1.1.002 | Initialize Go workspace with go.work | 1h | P0 | GO
Acceptance: go.work defines all modules, go mod tidy succeeds for all

Dependencies: 0.1.1.001

Risks: None

Deliverable: go.work file

0.1.1.003 | Create api/ directory with proto definitions | 2h | P0 | GO

Acceptance: Directory exists with VERSION file, buf.yaml configured

Dependencies: 0.1.1.001

Risks: None

Deliverable: api/ directory structure

0.1.1.004 | Initialize Zig build system | 2h | P0 | ZIG

Acceptance: build.zig compiles hello-world, cross-compilation works for x86_64 and aarch64

Dependencies: 0.1.1.001

Risks: Zig version compatibility

Deliverable: build.zig with cross-compilation targets

0.1.1.005 | Create C/C++ CMake setup for GPU components | 2h | P0 | C

Acceptance: CMake builds on Linux x86_64, finds CUDA/ROCM if available

Dependencies: 0.1.1.001

Risks: CMake version differences across platforms

Deliverable: CMakeLists.txt with platform detection

0.1.1.006 | Set up Docker Compose development environment | 2h | P0 | OPS

Acceptance: docker-compose up brings up all dependencies (Postgres, Redis, etcd, NATS, Kafka, Grafana, Prometheus)

Dependencies: 0.1.1.001

Risks: Port conflicts on developer machines

Deliverable: docker-compose.yml

0.1.1.007 | Create Makefile with common tasks | 1h | P0 | BASH

Acceptance: make help lists all targets, make dev starts environment, make test runs tests

Dependencies: 0.1.1.002, 0.1.1.004, 0.1.1.005

Risks: None

Deliverable: Makefile with 20+ targets

0.1.1.008 | Set up buf for Protocol Buffer management | 1h | P0 | GO

Acceptance: buf generate produces Go + gRPC code from .proto files

Dependencies: 0.1.1.003

Risks: None

Deliverable: buf.yaml, buf.gen.yaml

0.1.1.009 | Create .gitignore and .gitattributes | 0.5h | P0 | BASH

Acceptance: All generated files, binaries, secrets ignored

Dependencies: None

Risks: None

Deliverable: .gitignore, .gitattributes

0.1.1.010 | Set up VERSION and CHANGELOG system | 1h | P0 | BASH

Acceptance: VERSION file exists, CHANGELOG.md with Keep a Changelog format

Dependencies: None

Risks: None

0.1.2 CI/CD Pipeline

0.1.2.001 | Create GitHub Actions workflow for Go build | 2h | P0 | OPS

Acceptance: Workflow triggers on PR, runs go build, go test, go vet

Dependencies: 0.1.1.002

Risks: None

Deliverable: .github/workflows/go-build.yml

0.1.2.002 | Create GitHub Actions workflow for Zig build | 2h | P0 | OPS

Acceptance: Workflow builds Zig components, runs Zig tests

Dependencies: 0.1.1.004

Risks: Zig toolchain availability in CI

Deliverable: .github/workflows/zig-build.yml

0.1.2.003 | Create GitHub Actions workflow for C/C++ build | 2h | P0 | OPS

Acceptance: Workflow builds CUDA/ROCm components (if available), runs C tests

Dependencies: 0.1.1.005

Risks: GPU runners may not be available

Deliverable: .github/workflows/cc-build.yml

0.1.2.004 | Set up code coverage reporting (Codecov) | 1h | P1 | OPS

Acceptance: Coverage reported on every PR, badge in README

Dependencies: 0.1.2.001

Risks: None

Deliverable: Codecov integration

0.1.2.005 | Set up static analysis (golangci-lint, zigtidy) | 2h | P1 | OPS

Acceptance: golangci-lint runs 20+ linters, zigtidy runs on Zig code

Dependencies: 0.1.2.001, 0.1.2.002

Risks: None

Deliverable: .golangci.yml, lint workflow

0.1.2.006 | Set up Dependabot for dependency updates | 1h | P2 | OPS

Acceptance: Dependabot creates PRs for Go, Zig, Docker updates

Dependencies: 0.1.1.001

Risks: None

Deliverable: .github/dependabot.yml

0.1.2.007 | Create release automation workflow | 2h | P1 | OPS

Acceptance: Tag push creates release with binaries for all platforms

Dependencies: 0.1.2.001, 0.1.2.002, 0.1.2.003

Risks: Cross-compilation issues

Deliverable: .github/workflows/release.yml

0.1.2.008 | Set up Docker image build and push | 2h | P1 | OPS

Acceptance: Multi-arch Docker images (amd64, arm64) built and pushed to registry

Dependencies: 0.1.1.006

Risks: None

Deliverable: Dockerfile, image push workflow

0.1.2.009 | Create pre-commit hooks | 1h | P1 | BASH
Acceptance: Hooks run lint, format, test before commit
Dependencies: 0.1.2.005
Risks: None
Deliverable: .pre-commit-config.yaml

0.1.2.010 | Set up ArgoCD for GitOps deployment | 2h | P2 | OPS
Acceptance: ArgoCD watches repo, auto-deploys to dev environment
Dependencies: 0.1.2.008
Risks: Requires K8s cluster
Deliverable: ArgoCD Application manifests

0.1.3 Development Environment

0.1.3.001 | Create Tiltfile for live development | 2h | P1 | OPS
Acceptance: tilt up builds and watches all services, auto-reloads on change
Dependencies: 0.1.1.006
Risks: None
Deliverable: Tiltfile

0.1.3.002 | Create dev container configuration | 2h | P2 | OPS
Acceptance: Dev container opens with all tools pre-installed
Dependencies: 0.1.1.006
Risks: None
Deliverable: .devcontainer/devcontainer.json

0.1.3.003 | Create local k3d/kind cluster setup | 2h | P1 | OPS
Acceptance: make k8s-up creates local cluster, deploys all services
Dependencies: 0.1.1.006
Risks: Resource requirements (8GB+ RAM)
Deliverable: scripts/k8s-setup.sh

0.1.3.004 | Create development documentation | 2h | P0 | DOC
Acceptance: DEVELOPMENT.md explains how to build, test, run
Dependencies: 0.1.1.007
Risks: None
Deliverable: DEVELOPMENT.md

0.1.3.005 | Set up hot reload for Go services | 1h | P1 | GO
Acceptance: air or equivalent reloads Go services on file change
Dependencies: 0.1.3.001
Risks: None
Deliverable: .air.toml

0.1.3.006 | Create seed data for development | 2h | P1 | DB
Acceptance: make seed populates database with realistic test data
Dependencies: 0.1.6.003 (Postgres setup)
Risks: None
Deliverable: scripts/seed-data.sql

0.1.3.007 | Create integration test harness | 3h | P0 | QA
Acceptance: Testcontainers-Go spins up dependencies, runs integration tests
Dependencies: 0.1.1.006
Risks: Docker-in-Docker for CI
Deliverable: test/harness.go

0.1.3.008 | Set up tracing (Jaeger) for development | 1h | P1 | OPS
Acceptance: All service calls traced, visible in Jaeger UI
Dependencies: 0.1.1.006
Risks: None
Deliverable: Jaeger in docker-compose.yml

0.1.3.009 | Create make targets for common workflows | 1h | P1 | BASH
Acceptance: make test, make test-integration, make benchmark, make lint all work
Dependencies: 0.1.1.007
Risks: None
Deliverable: Updated Makefile

0.1.3.010 | Document debugging procedures | 1h | P1 | DOC
Acceptance: DEBUGGING.md with common issues and solutions
Dependencies: 0.1.3.004
Risks: None
Deliverable: DEBUGGING.md

0.1.4 Code Standards & Tooling

0.1.4.001 | Create Go coding standards document | 1h | P0 | DOC
Acceptance: CODING_STANDARDS_GO.md with style guide, patterns, anti-patterns
Dependencies: None
Risks: None
Deliverable: CODING_STANDARDS_GO.md

0.1.4.002 | Create Zig coding standards document | 1h | P0 | DOC
Acceptance: CODING_STANDARDS_ZIG.md with allocator patterns, error handling
Dependencies: None
Risks: None
Deliverable: CODING_STANDARDS_ZIG.md

0.1.4.003 | Create C coding standards document | 1h | P0 | DOC
Acceptance: CODING_STANDARDS_C.md with memory safety, CUDA patterns
Dependencies: None
Risks: None
Deliverable: CODING_STANDARDS_C.md

0.1.4.004 | Configure gofumpt for Go formatting | 0.5h | P1 | GO
Acceptance: gofumpt runs in CI, stricter than gofmt
Dependencies: 0.1.2.001
Risks: None
Deliverable: .github/workflows/format.yml

0.1.4.005 | Set up goimports automation | 0.5h | P1 | GO

Acceptance: Imports sorted automatically in CI
Dependencies: 0.1.4.004
Risks: None
Deliverable: Part of format workflow

0.1.4.006 | Configure zig fmt automation | 0.5h | P1 | ZIG
Acceptance: zig fmt runs in CI, fails on unformatted code
Dependencies: 0.1.2.002
Risks: None
Deliverable: Part of Zig workflow

0.1.4.007 | Set up clang-format for C/C++ | 0.5h | P1 | C
Acceptance: clang-format runs in CI
Dependencies: 0.1.2.003
Risks: None
Deliverable: .clang-format

0.1.4.008 | Create architecture decision record (ADR) template | 1h | P0 | DOC
Acceptance: ADR template in docs/adr/, first ADR for monorepo decision
Dependencies: None
Risks: None
Deliverable: docs/adr/0001-use-monorepo.md, docs/adr/TEMPLATE.md

0.1.4.009 | Create pull request template | 0.5h | P0 | DOC
Acceptance: Template enforces checklist: tests, docs, ADR if needed
Dependencies: None
Risks: None
Deliverable: .github/pull_request_template.md

0.1.4.010 | Set up issue templates | 0.5h | P1 | DOC
Acceptance: Bug report, feature request, ADR proposal templates
Dependencies: None
Risks: None
Deliverable: .github/ISSUE_TEMPLATE/

0.2 CORE LIBRARIES (Weeks 1-2)

0.2.1 Go Shared Libraries

0.2.1.001 | Create pkg/errors: structured error handling | 3h | P0 | G0
Acceptance: Error wrapping, code categorization, stack traces, gRPC status mapping
Dependencies: None
Risks: None
Deliverable: pkg/errors/

0.2.1.002 | Create pkg/log: structured logging (slog-based) | 2h | P0 | G0
Acceptance: JSON/TTY output, level control, context propagation, sampling
Dependencies: None
Risks: None
Deliverable: pkg/log/

0.2.1.003 | Create pkg/config: configuration management | 3h | P0 | G0
Acceptance: Viper-based, env vars, files, defaults, validation, hot reload
Dependencies: None
Risks: None
Deliverable: pkg/config/

0.2.1.004 | Create pkg/crypto: encryption helpers | 2h | P0 | G0
Acceptance: AES-GCM, ChaCha20-Poly1305, key derivation (Argon2), secure random
Dependencies: None
Risks: None
Deliverable: pkg/crypto/

0.2.1.005 | Create pkg/netutil: network utilities | 2h | P0 | G0
Acceptance: Interface discovery, IP validation, port availability, NAT detection
Dependencies: None
Risks: None
Deliverable: pkg/netutil/

0.2.1.006 | Create pkg/retry: retry mechanisms | 2h | P0 | G0
Acceptance: Exponential backoff, jitter, context cancellation, custom predicates
Dependencies: None
Risks: None
Deliverable: pkg/retry/

0.2.1.007 | Create pkg/backoff: advanced backoff strategies | 2h | P1 | G0
Acceptance: Linear, exponential, decorrelated jitter, circuit breaker integration
Dependencies: 0.2.1.006
Risks: None
Deliverable: pkg/backoff/

0.2.1.008 | Create pkg/context: context utilities | 1h | P0 | G0
Acceptance: Context with values, deadlines, cancellation propagation
Dependencies: None
Risks: None
Deliverable: pkg/context/

0.2.1.009 | Create pkg/health: health check framework | 2h | P0 | G0
Acceptance: Liveness/readiness probes, dependency health, custom checks
Dependencies: None
Risks: None
Deliverable: pkg/health/

0.2.1.010 | Create pkg/metrics: Prometheus metrics helpers | 2h | P0 | G0
Acceptance: Counter, gauge, histogram, summary with labels, automatic exposition
Dependencies: None
Risks: None
Deliverable: pkg/metrics/

0.2.1.011 | Create pkg/tracing: OpenTelemetry tracing | 2h | P1 | G0
Acceptance: Span creation, context propagation, Jaeger export
Dependencies: None

Risks: None

Deliverable: pkg/tracing/

0.2.1.012 | Create pkg/validator: input validation | 2h | P0 | GO

Acceptance: Struct tag validation, custom validators, i18n error messages

Dependencies: None

Risks: None

Deliverable: pkg/validator/

0.2.1.013 | Create pkg/jwt: JWT handling | 2h | P1 | GO

Acceptance: Sign, verify, refresh tokens, SPIFFE-aware

Dependencies: 0.2.1.004

Risks: None

Deliverable: pkg/jwt/

0.2.1.014 | Create pkg/middleware: HTTP middleware collection | 3h | P0 | GO

Acceptance: Auth, logging, recovery, CORS, rate limit, request ID, compression

Dependencies: 0.2.1.002, 0.2.1.012

Risks: None

Deliverable: pkg/middleware/

0.2.1.015 | Create pkg/grpcutil: gRPC utilities | 2h | P0 | GO

Acceptance: Interceptors (auth, logging, retry, metrics), connection pooling

Dependencies: 0.2.1.014

Risks: None

Deliverable: pkg/grpcutil/

0.2.1.016 | Create pkg/websocket: WebSocket wrapper | 3h | P0 | GO

Acceptance: Connection management, heartbeat, reconnection, message framing

Dependencies: None

Risks: None

Deliverable: pkg/websocket/

0.2.1.017 | Create pkg/classads: ClassAds parser | 4h | P0 | GO

Acceptance: Parse HTCondor-style expressions, evaluate against attribute sets, all operators

Dependencies: 0.2.1.012

Risks: Expression complexity

Deliverable: pkg/classads/

0.2.1.018 | Create pkg/serde: serialization abstraction | 2h | P0 | GO

Acceptance: JSON, MessagePack, Cap'n Proto adapters with unified interface

Dependencies: None

Risks: None

Deliverable: pkg/serde/

0.2.1.019 | Create pkg/pubsub: publish-subscribe framework | 3h | P0 | GO

Acceptance: NATS and in-memory backends, typed channels, subscription management

Dependencies: 0.2.1.016

Risks: None

Deliverable: pkg/pubsub/

0.2.1.020 | Create pkg/discovery: service discovery client | 2h | P0 | G0
 Acceptance: etcd-based service registration and lookup, health-aware selection
 Dependencies: None
 Risks: None
 Deliverable: pkg/discovery/

0.2.1.021 | Create pkg/leader: leader election | 2h | P0 | G0
 Acceptance: etcd-based leader election, graceful failover, leader callbacks
 Dependencies: None
 Risks: None
 Deliverable: pkg/leader/

0.2.1.022 | Create pkg/workerpool: generic worker pool | 2h | P0 | G0
 Acceptance: Fixed/dynamic sizing, task submission, graceful shutdown, metrics
 Dependencies: None
 Risks: None
 Deliverable: pkg/workerpool/

0.2.1.023 | Create pkg/semaphore: weighted semaphore | 1h | P0 | G0
 Acceptance: Dynamic capacity, fair queuing, timeout support
 Dependencies: None
 Risks: None
 Deliverable: pkg/semaphore/

0.2.1.024 | Create pkg/lru: LRU cache implementation | 2h | P0 | G0
 Acceptance: O(1) operations, TTL support, size-based eviction, thread-safe
 Dependencies: None
 Risks: None
 Deliverable: pkg/lru/

0.2.1.025 | Create pkg/ratelimit: rate limiter | 2h | P0 | G0
 Acceptance: Token bucket, sliding window, per-key limits, Redis backend
 Dependencies: 0.2.1.024
 Risks: None
 Deliverable: pkg/ratelimit/

0.2.2 Zig System Libraries

0.2.2.001 | Create zig-serde: serialization library | 4h | P0 | ZIG
 Acceptance: Cap'n Proto message builder/reader, zero-copy where possible, cross-platform
 Dependencies: None
 Risks: Cap'n Proto spec complexity
 Deliverable: zig-serde/

0.2.2.002 | Create zig-net: network primitives | 3h | P0 | ZIG
 Acceptance: TCP/UDP sockets, non-blocking I/O, epoll/kqueue/IOCP abstraction
 Dependencies: None
 Risks: Platform differences (Linux/macOS/Windows)
 Deliverable: zig-net/

0.2.2.003 | Create zig-protocol: binary protocol framing | 3h | P0 | ZIG

Acceptance: Length-prefixed framing, version negotiation, checksum validation
Dependencies: 0.2.2.001
Risks: None
Deliverable: zig-protocol/

0.2.2.004 | Create zig-zeromq: ZeroMQ bindings | 4h | P0 | ZIG

Acceptance: ZMQ framing, socket types (DEALER, ROUTER, PUB, SUB), C interop
Dependencies: 0.2.2.002, 0.2.2.003
Risks: ZeroMQ C library compatibility
Deliverable: zig-zeromq/

0.2.2.005 | Create zig-compress: compression utilities | 2h | P1 | ZIG

Acceptance: zstd, lz4 bindings, streaming compression
Dependencies: None
Risks: None
Deliverable: zig-compress/

0.2.2.006 | Create zig-crypto: cryptographic primitives | 3h | P0 | ZIG

Acceptance: ChaCha20-Poly1305, X25519, Blake2b (for WireGuard), constant-time
Dependencies: None
Risks: Security-critical, needs audit
Deliverable: zig-crypto/

0.2.2.007 | Create zig-memory: memory management utilities | 2h | P0 | ZIG

Acceptance: Arena allocators, pool allocators, buffer management
Dependencies: None
Risks: None
Deliverable: zig-memory/

0.2.2.008 | Create zig-ring: lock-free ring buffer | 2h | P0 | ZIG

Acceptance: SPSC/MPSC variants, cache-line padding, batch operations
Dependencies: 0.2.2.007
Risks: Lock-free correctness
Deliverable: zig-ring/

0.2.2.009 | Create zig-pty: pseudo-terminal management | 3h | P0 | ZIG

Acceptance: PTY open/close, resize, signal forwarding, Linux/macOS support
Dependencies: None
Risks: Platform differences
Deliverable: zig-pty/

0.2.2.010 | Create zig-gpu: GPU detection and management | 4h | P0 | ZIG

Acceptance: Enumerate all GPU vendors, query memory/compute, NVML/ROCm-SMI parsing
Dependencies: None
Risks: Vendor API differences
Deliverable: zig-gpu/

0.2.3 C/C++ GPU Libraries

0.2.3.001 | Create cc-gpu-common: GPU abstraction headers | 2h | P0 | C

Acceptance: Unified device/memory/stream handles, vendor detection macros

Dependencies: None

Risks: None

Deliverable: cc-gpu-common/

0.2.3.002 | Create cc-cuda: NVIDIA CUDA backend | 4h | P0 | C

Acceptance: Device enumeration, memory alloc/free, kernel launch, stream sync, NVML metrics

Dependencies: 0.2.3.001

Risks: Requires NVIDIA GPU for testing

Deliverable: cc-cuda/

0.2.3.003 | Create cc-rocm: AMD ROCm backend | 4h | P0 | C

Acceptance: HIP device enumeration, memory management, kernel launch, rocSmiMetrics

Dependencies: 0.2.3.001

Risks: Requires AMD GPU for testing

Deliverable: cc-rocm/

0.2.3.004 | Create cc-oneapi: Intel oneAPI backend | 4h | P0 | C

Acceptance: Level Zero device enumeration, memory management, SYCL kernel launch

Dependencies: 0.2.3.001

Risks: Requires Intel GPU for testing

Deliverable: cc-oneapi/

0.2.3.005 | Create cc-mlx: Apple MLX backend | 4h | P0 | C

Acceptance: MLX device detection, unified memory allocation, Metal command queue

Dependencies: 0.2.3.001

Risks: Requires Apple Silicon for testing

Deliverable: cc-mlx/

0.2.3.006 | Create cc-sycl: Cross-platform SYCL runtime | 3h | P1 | C

Acceptance: Device selection, USM memory, ND-range kernels, Intel/AMD/NVIDIA

Dependencies: 0.2.3.001

Risks: SYCL implementation availability

Deliverable: cc-sycl/

0.2.3.007 | Create cc-interpose: LD_PRELOAD interception | 3h | P1 | C

Acceptance: Intercept CUDA calls, forward to remote, HAMI-compatible

Dependencies: 0.2.3.002

Risks: API compatibility across CUDA versions

Deliverable: cc-interpose/

0.2.3.008 | Create cc-metrics: GPU metrics collection | 2h | P0 | C

Acceptance: Temperature, utilization, memory, power, ECC errors, per-process stats

Dependencies: 0.2.3.001

Risks: Vendor API differences

Deliverable: cc-metrics/

0.2.3.009 | Create cc-compress: GPU compression | 2h | P2 | C

Acceptance: NVIDIA nvCOMP integration, GPU-accelerated compression

Dependencies: 0.2.3.002

Risks: None

Deliverable: cc-compress/

0.2.3.010 | Create cc-network: GPU RDMA helpers | 2h | P2 | C
Acceptance: GPUDirect RDMA registration, NIC-GPU memory mapping
Dependencies: 0.2.3.002
Risks: Requires InfiniBand/RoCE hardware
Deliverable: cc-network/

0.2.4 Infrastructure Services

0.2.4.001 | Deploy PostgreSQL 16 with HA setup | 2h | P0 | DB
Acceptance: Primary-replica streaming replication, automatic failover (Patroni)
Dependencies: 0.1.1.006
Risks: None
Deliverable: PostgreSQL in docker-compose.yml

0.2.4.002 | Deploy Redis Cluster 7 | 2h | P0 | DB
Acceptance: 3 master + 3 replica cluster, automatic failover, persistence enabled
Dependencies: 0.1.1.006
Risks: None
Deliverable: Redis Cluster in docker-compose.yml

0.2.4.003 | Deploy etcd cluster | 2h | P0 | DB
Acceptance: 3-node etcd cluster, TLS enabled, snapshot backup
Dependencies: 0.1.1.006
Risks: None
Deliverable: etcd in docker-compose.yml

0.2.4.004 | Deploy NATS + JetStream | 1h | P0 | OPS
Acceptance: JetStream enabled, streams created, cluster mode
Dependencies: 0.1.1.006
Risks: None
Deliverable: NATS in docker-compose.yml

0.2.4.005 | Deploy Apache Kafka 4.0 (KRaft) | 2h | P0 | OPS
Acceptance: 3-node KRaft cluster, topics auto-created, retention configured
Dependencies: 0.1.1.006
Risks: Memory requirements
Deliverable: Kafka in docker-compose.yml

0.2.4.006 | Deploy RabbitMQ | 1h | P0 | OPS
Acceptance: Management UI accessible, clustering enabled
Dependencies: 0.1.1.006
Risks: None
Deliverable: RabbitMQ in docker-compose.yml

0.2.4.007 | Deploy Prometheus + Grafana | 2h | P0 | OPS
Acceptance: Prometheus scraping, Grafana dashboards, alerting rules
Dependencies: 0.1.1.006
Risks: None
Deliverable: Prometheus/Grafana in docker-compose.yml

0.2.4.008 | Deploy Ceph (demo mode) | 2h | P1 | OPS
Acceptance: CephFS mountable, RGW accessible, 3 OSDs in docker-compose
Dependencies: 0.1.1.006
Risks: High resource usage
Deliverable: Ceph in docker-compose.yml

0.2.4.009 | Deploy HashiCorp Vault | 1h | P1 | OPS
Acceptance: Vault unsealed, KV engine mounted, policies defined
Dependencies: 0.1.1.006
Risks: None
Deliverable: Vault in docker-compose.yml

0.2.4.010 | Deploy Jaeger for tracing | 1h | P1 | OPS
Acceptance: All-in-one mode, UI accessible, trace ingestion
Dependencies: 0.1.1.006
Risks: None
Deliverable: Jaeger in docker-compose.yml

0.3 PROTOCOL DEFINITIONS (Week 2)

0.3.1 gRPC Service Definitions

0.3.1.001 | Define NodeService protobuf | 2h | P0 | G0
Acceptance: node.proto with Join, Heartbeat, Leave, GetNode, ListNodes, WatchNodes
Dependencies: 0.1.1.008
Risks: None
Deliverable: api/v1/node.proto

0.3.1.002 | Define SessionService protobuf | 2h | P0 | G0
Acceptance: session.proto with Create, Attach, Detach, Terminate, Migrate, SendInput, ResizePty
Dependencies: 0.3.1.001
Risks: None
Deliverable: api/v1/session.proto

0.3.1.003 | Define SchedulerService protobuf | 2h | P0 | G0
Acceptance: scheduler.proto with Schedule, Cancel, GetPool, Reserve, Release, WatchPool
Dependencies: 0.3.1.001
Risks: None
Deliverable: api/v1/scheduler.proto

0.3.1.004 | Define HealthService protobuf | 1h | P0 | G0
Acceptance: health.proto with GetClusterHealth, GetNodeHealth, StreamHealth, PredictFailures
Dependencies: 0.3.1.001
Risks: None
Deliverable: api/v1/health.proto

0.3.1.005 | Define AdvisoryService protobuf | 1h | P0 | G0
Acceptance: advisory.proto with ListAdvisories, Approve, Reject, GetExplanation
Dependencies: 0.3.1.001
Risks: None
Deliverable: api/v1/advisory.proto

0.3.1.006 | Define SecurityService protobuf | 1h | P0 | GO
Acceptance: security.proto with Authenticate, Authorize, RotateCredentials, AuditLog
Dependencies: 0.3.1.001
Risks: None
Deliverable: api/v1/security.proto

0.3.1.007 | Define BuildService protobuf | 1h | P0 | GO
Acceptance: build.proto with SubmitJob, CancelJob, GetStatus, StreamLogs, GetArtifacts
Dependencies: 0.3.1.003
Risks: None
Deliverable: api/v1/build.proto

0.3.1.008 | Define common types protobuf | 2h | P0 | GO
Acceptance: types.proto with Node, Session, Resource, GPU, HealthScore, Capability
Dependencies: 0.3.1.001
Risks: None
Deliverable: api/v1/types.proto

0.3.1.009 | Generate Go stubs from protobufs | 1h | P0 | GO
Acceptance: buf generate produces .pb.go files, compiles without errors
Dependencies: 0.3.1.001-0.3.1.008
Risks: None
Deliverable: Generated Go files

0.3.1.010 | Generate Zig bindings from protobufs | 2h | P0 | ZIG
Acceptance: Protobuf structs usable from Zig, compatible with Go wire format
Dependencies: 0.3.1.009
Risks: Zig protobuf library maturity
Deliverable: Generated Zig bindings

0.3.2 Event Schemas

0.3.2.001 | Define node event schema (Avro) | 1h | P0 | GO
Acceptance: NodeJoined, NodeLeft, NodeFailed, ResourcesChanged events
Dependencies: 0.3.1.008
Risks: None
Deliverable: schemas/node-events.avsc

0.3.2.002 | Define session event schema (Avro) | 1h | P0 | GO
Acceptance: SessionCreated, SessionTerminated, SessionMigrated, PaneResized events
Dependencies: 0.3.2.001
Risks: None
Deliverable: schemas/session-events.avsc

0.3.2.003 | Define scheduler event schema (Avro) | 1h | P0 | GO
Acceptance: JobScheduled, JobPreempted, ResourcesReserved, BindingChanged events
Dependencies: 0.3.2.001
Risks: None
Deliverable: schemas/scheduler-events.avsc

0.3.2.004 | Define audit event schema (Avro) | 1h | P0 | GO
 Acceptance: All CRUD operations, authentication, authorization events
 Dependencies: 0.3.2.001
 Risks: None
 Deliverable: schemas/audit-events.avsc

0.3.2.005 | Create Kafka topic definitions | 1h | P0 | OPS
 Acceptance: All topics created with correct partitions, replication, retention
 Dependencies: 0.3.2.001-0.3.2.004
 Risks: None
 Deliverable: scripts/kafka-topics.sh

0.3.2.006 | Create NATS stream definitions | 1h | P0 | OPS
 Acceptance: All streams created with correct subjects, retention, replication
 Dependencies: 0.3.2.001-0.3.2.004
 Risks: None
 Deliverable: scripts/nats-streams.sh

0.3.2.007 | Create event serialization library | 2h | P0 | GO
 Acceptance: Avro encode/decode for all event types, schema registry compatible
 Dependencies: 0.3.2.001-0.3.2.004
 Risks: None
 Deliverable: pkg/events/

0.3.2.008 | Create event publisher | 2h | P0 | GO
 Acceptance: Publish to Kafka (audit) or NATS (control) based on event type
 Dependencies: 0.3.2.007
 Risks: None
 Deliverable: pkg/events/publisher.go

0.3.2.009 | Create event consumer framework | 2h | P0 | GO
 Acceptance: Consumer groups, auto-rebalance, offset management, error handling
 Dependencies: 0.3.2.007
 Risks: None
 Deliverable: pkg/events/consumer.go

0.3.2.010 | Create event replay capability | 2h | P1 | GO
 Acceptance: Replay events from offset, time-based replay, deterministic
 Dependencies: 0.3.2.008, 0.3.2.009
 Risks: None
 Deliverable: pkg/events/replay.go

0.4 DATABASE SETUP (Week 2-3)

0.4.1 PostgreSQL Schema Implementation

0.4.1.001 | Create migration framework (golang-migrate) | 2h | P0 | DB
 Acceptance: Up/down migrations, version tracking, rollback support
 Dependencies: 0.2.4.001
 Risks: None
 Deliverable: migrations/ directory, migrate CLI

0.4.1.002 | Migration 001: Create nodes table | 1h | P0 | DB
Acceptance: Table created with all columns, indexes, triggers
Dependencies: 0.4.1.001
Risks: None
Deliverable: migrations/001_create_nodes.up.sql

0.4.1.003 | Migration 002: Create gpu_devices table | 1h | P0 | DB
Acceptance: Table with foreign key to nodes, indexes
Dependencies: 0.4.1.002
Risks: None
Deliverable: migrations/002_create_gpu_devices.up.sql

0.4.1.004 | Migration 003: Create sessions table | 1h | P0 | DB
Acceptance: Table with all columns, indexes for owner/status/node
Dependencies: 0.4.1.002
Risks: None
Deliverable: migrations/003_create_sessions.up.sql

0.4.1.005 | Migration 004: Create session_windows table | 1h | P0 | DB
Acceptance: Table with foreign key to sessions
Dependencies: 0.4.1.004
Risks: None
Deliverable: migrations/004_create_session_windows.up.sql

0.4.1.006 | Migration 005: Create session_panes table | 1h | P0 | DB
Acceptance: Table with foreign keys to windows and nodes
Dependencies: 0.4.1.005
Risks: None
Deliverable: migrations/005_create_session_panes.up.sql

0.4.1.007 | Migration 006: Create reservations table | 1h | P0 | DB
Acceptance: Table with indexes for session, node, status
Dependencies: 0.4.1.002, 0.4.1.004
Risks: None
Deliverable: migrations/006_create_reservations.up.sql

0.4.1.008 | Migration 007: Create migration_history table | 1h | P0 | DB
Acceptance: Table recording all session migrations
Dependencies: 0.4.1.004
Risks: None
Deliverable: migrations/007_create_migration_history.up.sql

0.4.1.009 | Migration 008: Create audit_log table with partitioning | 2h | P0 | DB
Acceptance: Partitioned table, monthly partitions, indexes
Dependencies: 0.4.1.001
Risks: None
Deliverable: migrations/008_create_audit_log.up.sql

0.4.1.010 | Migration 009: Create users table | 1h | P0 | DB
Acceptance: Table with SPIFFE ID unique index

Dependencies: 0.4.1.001

Risks: None

Deliverable: migrations/009_create_users.up.sql

0.4.1.011 | Migration 010: Create health_snapshots table | 1h | P0 | DB

Acceptance: Table with indexes for node and time

Dependencies: 0.4.1.002

Risks: None

Deliverable: migrations/010_create_health_snapshots.up.sql

0.4.1.012 | Migration 011: Create llm_advisories table | 1h | P0 | DB

Acceptance: Table with indexes for status, type, risk

Dependencies: 0.4.1.001

Risks: None

Deliverable: migrations/011_create_llm_advisories.up.sql

0.4.1.013 | Create auto-partitioning function for audit_log | 2h | P0 | DB

Acceptance: Automatically creates new monthly partitions

Dependencies: 0.4.1.009

Risks: None

Deliverable: migrations/012_audit_auto_partition.up.sql

0.4.1.014 | Create audit trigger function | 1h | P0 | DB

Acceptance: Trigger on all tables writes to audit_log

Dependencies: 0.4.1.009

Risks: Performance impact

Deliverable: migrations/013_audit_trigger.up.sql

0.4.1.015 | Create update_updated_at trigger function | 1h | P0 | DB

Acceptance: All tables with updated_at have auto-update trigger

Dependencies: 0.4.1.002-0.4.1.012

Risks: None

Deliverable: migrations/014_updated_at_trigger.up.sql

0.4.1.016 | Create database seed script for development | 2h | P1 | DB

Acceptance: Populates realistic test data for all tables

Dependencies: 0.4.1.015

Risks: None

Deliverable: scripts/seed-data.sql

0.4.1.017 | Create database connection pool manager | 2h | P0 | GO

Acceptance: Connection pooling, retry, health check, metrics

Dependencies: 0.2.1.006, 0.4.1.001

Risks: None

Deliverable: internal/db/pool.go

0.4.1.018 | Create database transaction manager | 2h | P0 | GO

Acceptance: Nested transactions, savepoints, automatic rollback on panic

Dependencies: 0.4.1.017

Risks: None

Deliverable: internal/db/tx.go

0.4.1.019 | Create query builder (squirrel-based) | 2h | P0 | GO

Acceptance: Type-safe query building for all entities

Dependencies: 0.4.1.017

Risks: None

Deliverable: internal/db/query.go

0.4.1.020 | Create database health check | 1h | P0 | GO

Acceptance: Reports connection pool stats, query latency

Dependencies: 0.4.1.017

Risks: None

Deliverable: internal/db/health.go

0.4.2 etcd Schema Implementation

0.4.2.001 | Create etcd key structure constants | 1h | P0 | GO

Acceptance: All key prefixes defined as constants, hierarchical

Dependencies: 0.2.4.003

Risks: None

Deliverable: pkg/etcd/keys.go

0.4.2.002 | Create etcd client wrapper | 2h | P0 | GO

Acceptance: Connection pooling, retry, watch, lease management

Dependencies: 0.2.4.003

Risks: None

Deliverable: pkg/etcd/client.go

0.4.2.003 | Create etcd serialization helpers | 1h | P0 | GO

Acceptance: JSON encode/decode with version field for migrations

Dependencies: 0.4.2.002

Risks: None

Deliverable: pkg/etcd/serde.go

0.4.2.004 | Create etcd watch framework | 2h | P0 | GO

Acceptance: Watch with prefix, filter, handler registration

Dependencies: 0.4.2.002

Risks: None

Deliverable: pkg/etcd/watch.go

0.4.2.005 | Create etcd lease manager | 2h | P0 | GO

Acceptance: Automatic lease renewal, TTL management, cleanup

Dependencies: 0.4.2.002

Risks: None

Deliverable: pkg/etcd/lease.go

0.4.2.006 | Create etcd distributed lock | 2h | P0 | GO

Acceptance: Lock/unlock with timeout, fairness, automatic release

Dependencies: 0.4.2.005

Risks: Clock skew

Deliverable: pkg/etcd/lock.go

0.4.2.007 | Create etcd election helper | 2h | P0 | G0
Acceptance: Campaign, observe, resign with callbacks
Dependencies: 0.4.2.006
Risks: None
Deliverable: pkg/etcd/election.go

0.4.2.008 | Create etcd transaction helpers | 2h | P0 | G0
Acceptance: Compare-and-swap, multi-key transactions
Dependencies: 0.4.2.002
Risks: None
Deliverable: pkg/etcd/tx.go

0.4.2.009 | Implement node registry in etcd | 2h | P0 | G0
Acceptance: CRUD operations for Node, watch for changes
Dependencies: 0.4.2.001-0.4.2.008
Risks: None
Deliverable: internal/registry/nodes.go

0.4.2.010 | Implement session registry in etcd | 2h | P0 | G0
Acceptance: CRUD operations for Session, routing table management
Dependencies: 0.4.2.009
Risks: None
Deliverable: internal/registry/sessions.go

0.4.3 Redis Schema Implementation

0.4.3.001 | Create Redis client wrapper | 2h | P0 | G0
Acceptance: Cluster-aware, connection pooling, pipeline support
Dependencies: 0.2.4.002
Risks: None
Deliverable: pkg/redis/client.go

0.4.3.002 | Create session state cache | 2h | P0 | G0
Acceptance: JSON serialization, TTL management, CRDT vector clock
Dependencies: 0.4.3.001
Risks: None
Deliverable: internal/cache/sessions.go

0.4.3.003 | Create resource pool cache | 2h | P0 | G0
Acceptance: Atomic updates, pub/sub for changes
Dependencies: 0.4.3.001
Risks: None
Deliverable: internal/cache/resources.go

0.4.3.004 | Create GPU status cache | 1h | P0 | G0
Acceptance: Per-GPU status, metrics, utilization
Dependencies: 0.4.3.001
Risks: None
Deliverable: internal/cache/gpus.go

0.4.3.005 | Create rate limiter cache | 2h | P0 | G0

Acceptance: Token bucket per key, sliding window, Redis Lua scripts
Dependencies: 0.4.3.001
Risks: None
Deliverable: internal/cache/ratelimit.go

0.4.3.006 | Create pub/sub channels | 1h | P0 | G0
Acceptance: Node events, session events, scheduler events
Dependencies: 0.4.3.001
Risks: None
Deliverable: internal/cache/pubsub.go

0.4.3.007 | Create health metrics cache | 1h | P0 | G0
Acceptance: Latest health snapshot per node, historical window
Dependencies: 0.4.3.001
Risks: None
Deliverable: internal/cache/health.go

0.4.3.008 | Create cache invalidation framework | 2h | P0 | G0
Acceptance: Event-driven invalidation, multi-level cache coherence
Dependencies: 0.4.3.002-0.4.3.007
Risks: Race conditions
Deliverable: internal/cache/invalidate.go

0.4.3.009 | Implement cache metrics | 1h | P1 | G0
Acceptance: Hit/miss rates, eviction counts, latency histograms
Dependencies: 0.4.3.001
Risks: None
Deliverable: internal/cache/metrics.go

0.4.3.010 | Create cache warming system | 2h | P1 | G0
Acceptance: Pre-populate hot data on startup, background refresh
Dependencies: 0.4.3.008
Risks: Startup time impact
Deliverable: internal/cache/warm.go

PHASE 1: CORE INFRASTRUCTURE (Weeks 5-12)

1.1 NODE DISCOVERY SERVICE (Weeks 5-6)

1.1.1 SWIM Gossip Protocol

1.1.1.001 | Implement SWIM message types | 3h | P0 | G0
Acceptance: Ping, PingReq, Ack, Suspect, Alive, Dead message structs
Dependencies: 0.3.1.001, 0.2.1.001
Risks: None
Deliverable: internal/gossip/message.go

1.1.1.002 | Implement UDP transport for gossip | 3h | P0 | G0
Acceptance: Send/receive gossip messages over UDP, checksum validation
Dependencies: 1.1.1.001

Risks: Firewall blocking UDP

Deliverable: internal/gossip/transport.go

1.1.1.003 | Implement gossip protocol state machine | 4h | P0 | G0

Acceptance: Protocol state transitions, member list management

Dependencies: 1.1.1.002

Risks: State machine correctness

Deliverable: internal/gossip/protocol.go

1.1.1.004 | Implement failure detection (phi accrual) | 3h | P0 | G0

Acceptance: Phi threshold configuration, adaptive intervals, false positive rate < 1%

Dependencies: 1.1.1.003

Risks: Tuning sensitivity

Deliverable: internal/gossip/detector.go

1.1.1.005 | Implement indirect ping (suspect handling) | 2h | P0 | G0

Acceptance: K random witnesses, suspect timeout handling

Dependencies: 1.1.1.003

Risks: None

Deliverable: internal/gossip/indirect.go

1.1.1.006 | Implement dissemination component | 2h | P0 | G0

Acceptance: Piggybacked updates, bounded broadcast buffer

Dependencies: 1.1.1.003

Risks: None

Deliverable: internal/gossip/dissemination.go

1.1.1.007 | Implement gossip compression | 2h | P1 | G0

Acceptance: Message batch compression, zstd, reduce bandwidth 50%+

Dependencies: 1.1.1.001

Risks: CPU overhead

Deliverable: internal/gossip/compress.go

1.1.1.008 | Implement gossip encryption | 2h | P0 | G0

Acceptance: AES-256-GCM for gossip messages, key rotation

Dependencies: 0.2.1.004

Risks: Performance impact

Deliverable: internal/gossip/crypto.go

1.1.1.009 | Implement gossip metrics | 1h | P1 | G0

Acceptance: Messages sent/received, bandwidth, false positives

Dependencies: 0.2.1.010

Risks: None

Deliverable: internal/gossip/metrics.go

1.1.1.010 | SWIM integration tests | 4h | P0 | QA

Acceptance: 3-node cluster simulation, failure injection, partition simulation

Dependencies: 1.1.1.001-1.1.1.009

Risks: Test flakiness due to timing

Deliverable: internal/gossip/protocol_test.go

1.1.2 Node Registration & Lifecycle

- 1.1.2.001 | Implement node join handler | 3h | P0 | GO
Acceptance: Validate join request, assign ID, persist to etcd, notify cluster
Dependencies: 0.4.2.009, 1.1.1.003
Risks: Race condition on simultaneous joins
Deliverable: internal/nodes/join.go
- 1.1.2.002 | Implement node fingerprinting (CPU) | 3h | P0 | GO
Acceptance: Detect architecture, cores, threads, features (AVX, NEON), cache sizes
Dependencies: None
Risks: Platform differences (x86_64, ARM64)
Deliverable: internal/nodes/fingerprint_cpu.go
- 1.1.2.003 | Implement node fingerprinting (Memory) | 2h | P0 | GO
Acceptance: Detect total RAM, speed, channels, NUMA topology
Dependencies: 1.1.2.002
Risks: None
Deliverable: internal/nodes/fingerprint_memory.go
- 1.1.2.004 | Implement node fingerprinting (GPU) | 3h | P0 | GO
Acceptance: Detect all GPU vendors, models, memory, compute capability
Dependencies: 0.2.3.001, 0.2.3.008
Risks: Requires GPU hardware for testing
Deliverable: internal/nodes/fingerprint_gpu.go
- 1.1.2.005 | Implement node fingerprinting (Storage) | 2h | P0 | GO
Acceptance: Detect disks, sizes, types (NVMe, SSD, HDD), filesystems
Dependencies: None
Risks: None
Deliverable: internal/nodes/fingerprint_storage.go
- 1.1.2.006 | Implement node fingerprinting (Network) | 2h | P0 | GO
Acceptance: Detect interfaces, speeds, MAC addresses, IP addresses
Dependencies: None
Risks: None
Deliverable: internal/nodes/fingerprint_network.go
- 1.1.2.007 | Implement capability advertisement | 2h | P0 | GO
Acceptance: Convert fingerprinted resources to Capability list
Dependencies: 1.1.2.002-1.1.2.006
Risks: None
Deliverable: internal/nodes/capabilities.go
- 1.1.2.008 | Implement node heartbeat handler | 2h | P0 | GO
Acceptance: Process heartbeat, update last_seen, check health score
Dependencies: 0.4.2.009, 1.1.1.003
Risks: None
Deliverable: internal/nodes/heartbeat.go
- 1.1.2.009 | Implement node leave handler | 2h | P0 | GO

Acceptance: Graceful departure, resource cleanup, session migration trigger
Dependencies: 1.1.2.001
Risks: Incomplete cleanup on crash
Deliverable: internal/nodes/leave.go

1.1.2.010 | Implement node failure handler | 3h | P0 | G0

Acceptance: Mark FAILED, trigger session evacuation, update resource pool
Dependencies: 1.1.1.004, 1.1.2.008
Risks: False positive failures
Deliverable: internal/nodes/failure.go

1.1.2.011 | Implement node role management | 2h | P0 | G0

Acceptance: WORKER, CONTROL, HYBRID roles, role transitions
Dependencies: 1.1.2.001
Risks: None
Deliverable: internal/nodes/roles.go

1.1.2.012 | Implement node labels system | 2h | P1 | G0

Acceptance: Key-value labels, selector queries, validation
Dependencies: 1.1.2.001
Risks: None
Deliverable: internal/nodes/labels.go

1.1.2.013 | Implement node version compatibility | 2h | P0 | G0

Acceptance: Check version on join, compatibility matrix, upgrade path
Dependencies: 1.1.2.001
Risks: Breaking changes between versions
Deliverable: internal/nodes/version.go

1.1.2.014 | Implement bootstrap node discovery | 3h | P0 | G0

Acceptance: mDNS discovery, manual bootstrap IP, rendezvous server
Dependencies: 0.2.1.005
Risks: mDNS may not work across subnets
Deliverable: internal/nodes/bootstrap.go

1.1.2.015 | Implement split-brain detection | 3h | P0 | G0

Acceptance: Quorum check, partition detection, automatic fencing
Dependencies: 0.4.2.006, 1.1.2.008
Risks: Complex distributed logic
Deliverable: internal/nodes/splitbrain.go

1.1.2.016 | Implement network partition recovery | 3h | P0 | G0

Acceptance: Automatic merge after partition, conflict resolution
Dependencies: 1.1.2.015
Risks: Data loss during merge
Deliverable: internal/nodes/partition.go

1.1.2.017 | Create node service gRPC handler | 3h | P0 | G0

Acceptance: All NodeService RPCs implemented
Dependencies: 0.3.1.001, 0.3.1.009, 1.1.2.001-1.1.2.016
Risks: None

Deliverable: internal/nodes/server.go

1.1.2.018 | Create node service HTTP handler | 3h | P0 | GO

Acceptance: All REST endpoints for nodes

Dependencies: 1.1.2.017

Risks: None

Deliverable: internal/nodes/http.go

1.1.2.019 | Create node event publisher | 2h | P0 | GO

Acceptance: Publish NodeJoined, NodeLeft, NodeFailed, ResourcesChanged events

Dependencies: 0.3.2.008

Risks: None

Deliverable: internal/nodes/events.go

1.1.2.020 | Node discovery integration tests | 4h | P0 | QA

Acceptance: 5-node cluster, join/leave/fail scenarios, partition recovery

Dependencies: 1.1.2.001-1.1.2.019

Risks: Test environment stability

Deliverable: test/nodes/integration_test.go

1.2 WIREGUARD MESH NETWORK (Weeks 6-7)

1.2.1 WireGuard Integration

1.2.1.001 | Create WireGuard Go bindings | 3h | P0 | GO

Acceptance: Device creation, peer management, key generation via wgctrl

Dependencies: None

Risks: WireGuard kernel module availability

Deliverable: internal/wireguard/device.go

1.2.1.002 | Implement key pair generation | 1h | P0 | GO

Acceptance: X25519 key pairs, persistent storage, rotation support

Dependencies: 1.2.1.001

Risks: None

Deliverable: internal/wireguard/keys.go

1.2.1.003 | Implement peer configuration | 3h | P0 | GO

Acceptance: Add/remove/update peers, allowed IPs, persistent keepalive

Dependencies: 1.2.1.001

Risks: None

Deliverable: internal/wireguard/peer.go

1.2.1.004 | Implement mesh topology formation | 4h | P0 | GO

Acceptance: Full mesh between all nodes, automatic peer addition on join

Dependencies: 1.2.1.003, 1.1.2.001

Risks: Quadratic growth of peers

Deliverable: internal/wireguard/mesh.go

1.2.1.005 | Implement subnet allocation | 2h | P0 | GO

Acceptance: Automatic WireGuard subnet (100.64.0.0/10), per-node IP

Dependencies: 1.2.1.004

Risks: IP conflicts

Deliverable: internal/wireguard/subnet.go

1.2.1.006 | Implement NAT traversal (STUN) | 3h | P0 | G0

Acceptance: Public IP discovery, NAT type detection

Dependencies: None

Risks: None

Deliverable: internal/wireguard/stun.go

1.2.1.007 | Implement UDP hole punching | 3h | P0 | G0

Acceptance: Direct connection through NAT, fallback to relay

Dependencies: 1.2.1.006

Risks: Symmetric NAT may fail

Deliverable: internal/wireguard/holepunch.go

1.2.1.008 | Implement DERP relay fallback | 3h | P1 | G0

Acceptance: Relay through control plane when direct fails

Dependencies: 1.2.1.007

Risks: Bandwidth through relay

Deliverable: internal/wireguard/relay.go

1.2.1.009 | Implement SSH tunnel fallback | 3h | P0 | G0

Acceptance: Reverse SSH tunnel when UDP blocked, automatic promotion to WireGuard

Dependencies: None

Risks: TCP-over-TCP issues

Deliverable: internal/wireguard/sshtunnel.go

1.2.1.010 | Implement connection health monitoring | 2h | P0 | G0

Acceptance: RTT measurement, packet loss detection, automatic reconnection

Dependencies: 1.2.1.004

Risks: None

Deliverable: internal/wireguard/health.go

1.2.1.011 | Implement automatic reconnection | 2h | P0 | G0

Acceptance: Detect disconnection, retry with backoff, circuit breaker

Dependencies: 1.2.1.010, 0.2.1.006

Risks: Reconnection storms

Deliverable: internal/wireguard/reconnect.go

1.2.1.012 | Implement multi-path routing | 3h | P2 | G0

Acceptance: Use multiple interfaces if available, bond for throughput

Dependencies: 1.2.1.004

Risks: Complex routing logic

Deliverable: internal/wireguard/multipath.go

1.2.1.013 | Implement WireGuard metrics | 1h | P1 | G0

Acceptance: Bytes transferred, peer latency, handshake timing

Dependencies: 0.2.1.010

Risks: None

Deliverable: internal/wireguard/metrics.go

1.2.1.014 | Implement fallback chain | 2h | P0 | G0
Acceptance: Direct → Hole Punch → Relay → SSH Tunnel, automatic promotion
Dependencies: 1.2.1.004-1.2.1.009
Risks: Fallback loops
Deliverable: internal/wireguard/fallback.go

1.2.1.015 | WireGuard integration tests | 4h | P0 | QA
Acceptance: 3-node mesh, NAT simulation, failover, reconnection
Dependencies: 1.2.1.001-1.2.1.014
Risks: Requires network namespace for testing
Deliverable: test/wireguard/integration_test.go

1.3 MESSAGING INFRASTRUCTURE (Week 7)

1.3.1 NATS Integration

1.3.1.001 | Create NATS connection manager | 2h | P0 | G0
Acceptance: Connection pooling, auto-reconnect, TLS
Dependencies: 0.2.4.004
Risks: None
Deliverable: internal/messaging/nats.go

1.3.1.002 | Implement JetStream publisher | 2h | P0 | G0
Acceptance: Publish with ack, deduplication, async batch
Dependencies: 1.3.1.001
Risks: None
Deliverable: internal/messaging/publisher.go

1.3.1.003 | Implement JetStream consumer | 2h | P0 | G0
Acceptance: Durable consumer, auto-ack/nack, backoff retry
Dependencies: 1.3.1.001
Risks: None
Deliverable: internal/messaging/consumer.go

1.3.1.004 | Implement request-reply pattern | 2h | P0 | G0
Acceptance: Request with timeout, reply correlation, scatter-gather
Dependencies: 1.3.1.001
Risks: Timeout tuning
Deliverable: internal/messaging/reqreply.go

1.3.1.005 | Create stream definitions | 1h | P0 | G0
Acceptance: All streams created with retention, replication
Dependencies: 0.3.2.006
Risks: None
Deliverable: internal/messaging/streams.go

1.3.1.006 | Implement dead letter handling | 2h | P0 | G0
Acceptance: Max retries, DLQ routing, alerting
Dependencies: 1.3.1.003
Risks: None
Deliverable: internal/messaging/deadletter.go

1.3.1.007 | Implement backpressure management | 2h | P0 | G0

Acceptance: Rate limiting, buffer management, flow control

Dependencies: 1.3.1.003

Risks: None

Deliverable: internal/messaging/backpressure.go

1.3.1.008 | NATS integration tests | 3h | P0 | QA

Acceptance: Pub/sub, req/reply, DLQ, backpressure scenarios

Dependencies: 1.3.1.001-1.3.1.007

Risks: None

Deliverable: test/messaging/integration_test.go

1.3.2 Kafka Integration

1.3.2.001 | Create Kafka producer | 2h | P0 | G0

Acceptance: Async producer, batching, compression, idempotency

Dependencies: 0.2.4.005

Risks: None

Deliverable: internal/messaging/kafka_producer.go

1.3.2.002 | Create Kafka consumer | 2h | P0 | G0

Acceptance: Consumer groups, manual offset, rebalancing

Dependencies: 1.3.2.001

Risks: Rebalance storms

Deliverable: internal/messaging/kafka_consumer.go

1.3.2.003 | Implement exactly-once processing | 3h | P0 | G0

Acceptance: Transactional producer, idempotent consumer

Dependencies: 1.3.2.001, 1.3.2.002

Risks: Complex coordination

Deliverable: internal/messaging/exactly_once.go

1.3.2.004 | Implement event sourcing | 3h | P0 | G0

Acceptance: Event append, snapshot, replay from offset

Dependencies: 1.3.2.002

Risks: Storage growth

Deliverable: internal/messaging/eventsource.go

1.3.2.005 | Kafka integration tests | 3h | P0 | QA

Acceptance: Producer/consumer, exactly-once, event replay

Dependencies: 1.3.2.001-1.3.2.004

Risks: None

Deliverable: test/messaging/kafka_test.go

1.4 API GATEWAY (Weeks 7-8)

1.4.1 HTTP Server

1.4.1.001 | Initialize Gin server with middleware stack | 2h | P0 | G0

Acceptance: Server starts, middleware chain executes in order

Dependencies: 0.2.1.014

Risks: None

Deliverable: internal/gateway/server.go

1.4.1.002 | Implement authentication middleware (mTLS) | 3h | P0 | G0

Acceptance: Extract SPIFFE ID from client cert, reject unauthenticated

Dependencies: 0.2.1.004

Risks: Certificate validation

Deliverable: internal/gateway/authn.go

1.4.1.003 | Implement authorization middleware (OPA) | 3h | P0 | G0

Acceptance: OPA policy evaluation, RBAC enforcement

Dependencies: 0.2.1.014

Risks: Policy evaluation performance

Deliverable: internal/gateway/authz.go

1.4.1.004 | Implement rate limiting middleware | 2h | P0 | G0

Acceptance: Per-user and global limits, sliding window

Dependencies: 0.2.1.025

Risks: None

Deliverable: internal/gateway/ratelimit.go

1.4.1.005 | Implement logging middleware | 2h | P0 | G0

Acceptance: Request/response logging, correlation IDs, timing

Dependencies: 0.2.1.002

Risks: None

Deliverable: internal/gateway/logging.go

1.4.1.006 | Implement recovery middleware | 1h | P0 | G0

Acceptance: Panic recovery, error response, stack trace logging

Dependencies: None

Risks: None

Deliverable: internal/gateway/recovery.go

1.4.1.007 | Implement CORS middleware | 1h | P1 | G0

Acceptance: Configurable origins, methods, headers

Dependencies: None

Risks: None

Deliverable: internal/gateway/cors.go

1.4.1.008 | Implement request ID middleware | 1h | P0 | G0

Acceptance: UUID generation, propagation to downstream services

Dependencies: None

Risks: None

Deliverable: internal/gateway/requestid.go

1.4.1.009 | Implement compression middleware | 1h | P1 | G0

Acceptance: Gzip/Brotli for responses > 1KB

Dependencies: None

Risks: CPU overhead

Deliverable: internal/gateway/compress.go

1.4.1.010 | Implement metrics middleware | 1h | P0 | GO
Acceptance: Request count, latency histogram, status code counters
Dependencies: 0.2.1.010
Risks: None
Deliverable: internal/gateway/metrics.go

1.4.1.011 | Create route registration system | 2h | P0 | GO
Acceptance: All service routes registered, versioned API paths
Dependencies: 1.4.1.001
Risks: None
Deliverable: internal/gateway/routes.go

1.4.1.012 | Implement WebSocket upgrade handler | 3h | P0 | GO
Acceptance: HTTP upgrade to WebSocket, subprotocol negotiation
Dependencies: 0.2.1.016
Risks: Connection limits
Deliverable: internal/gateway/websocket.go

1.4.1.013 | Create health check endpoint | 1h | P0 | GO
Acceptance: /healthz (liveness), /ready (readiness), dependency checks
Dependencies: 0.2.1.009
Risks: None
Deliverable: internal/gateway/health.go

1.4.1.014 | Create API documentation endpoint | 1h | P1 | GO
Acceptance: /docs serves OpenAPI spec, Swagger UI
Dependencies: 1.4.1.011
Risks: None
Deliverable: internal/gateway/docs.go

1.4.1.015 | Gateway integration tests | 4h | P0 | QA
Acceptance: All middleware, routing, auth, rate limiting tested
Dependencies: 1.4.1.001-1.4.1.014
Risks: None
Deliverable: test/gateway/integration_test.go

1.4.2 gRPC-Gateway

1.4.2.001 | Set up gRPC-Gateway | 2h | P0 | GO
Acceptance: REST endpoints proxy to gRPC services
Dependencies: 0.3.1.009
Risks: None
Deliverable: internal/gateway/grpc.go

1.4.2.002 | Configure HTTP annotation mapping | 2h | P0 | GO
Acceptance: All protobuf services have HTTP annotations
Dependencies: 0.3.1.001-0.3.1.007
Risks: None
Deliverable: Updated .proto files with annotations

1.4.2.003 | Implement gRPC load balancing | 2h | P0 | G0
Acceptance: Round-robin across service instances, health-aware
Dependencies: 0.2.1.015
Risks: None
Deliverable: internal/gateway/grpc_lb.go

1.4.2.004 | gRPC-Gateway integration tests | 2h | P0 | QA
Acceptance: REST calls correctly proxy to gRPC backends
Dependencies: 1.4.2.001-1.4.2.003
Risks: None
Deliverable: test/gateway/grpc_test.go

1.5 SECURITY MANAGER (Weeks 8-9)

1.5.1 SPIFFE/SPIRE Integration

1.5.1.001 | Create SPIFFE ID format and validation | 2h | P0 | G0
Acceptance: spiffe://cluster.trustdomain/node/{id} format, validation
Dependencies: None
Risks: None
Deliverable: internal/security/spiffe.go

1.5.1.002 | Implement node attestation (join-time) | 3h | P0 | G0
Acceptance: Challenge-response, proof-of-possession (pubkey)
Dependencies: 1.5.1.001, 1.2.1.002
Risks: None
Deliverable: internal/security/attestation.go

1.5.1.003 | Implement SVID issuance and rotation | 3h | P0 | G0
Acceptance: X.509 SVID, 24h TTL, automatic rotation at 80%
Dependencies: 1.5.1.002
Risks: Clock skew
Deliverable: internal/security/svid.go

1.5.1.004 | Implement workload identity propagation | 2h | P0 | G0
Acceptance: SPIFFE ID in gRPC metadata, HTTP headers
Dependencies: 1.5.1.003
Risks: None
Deliverable: internal/security/identity.go

1.5.1.005 | Create trust bundle management | 2h | P0 | G0
Acceptance: Bundle distribution, rotation, revocation
Dependencies: 1.5.1.003
Risks: None
Deliverable: internal/security/bundle.go

1.5.2 Access Control

1.5.2.001 | Integrate Open Policy Agent (OPA) | 3h | P0 | G0
Acceptance: OPA sidecar, policy evaluation, decision logging
Dependencies: None

Risks: OPA performance
Deliverable: internal/security/opa.go

1.5.2.002 | Define RBAC policies | 2h | P0 | G0

Acceptance: Admin, Operator, User roles, resource permissions
Dependencies: 1.5.2.001
Risks: None
Deliverable: policies/rbac.rego

1.5.2.003 | Define node-level policies | 2h | P0 | G0

Acceptance: Node can only modify own resources, cross-node restrictions
Dependencies: 1.5.2.001
Risks: None
Deliverable: policies/node.rego

1.5.2.004 | Implement policy hot-reload | 2h | P1 | G0

Acceptance: Policy changes without restart, validation
Dependencies: 1.5.2.001
Risks: Invalid policy syntax
Deliverable: internal/security/policy_reload.go

1.5.2.005 | Security manager integration tests | 4h | P0 | QA

Acceptance: Attestation, authz, policy evaluation, rotation
Dependencies: 1.5.1.001-1.5.2.004
Risks: Complex test setup
Deliverable: test/security/integration_test.go

1.6 CONTROL PLANE DEPLOYMENT (Weeks 9-10)

1.6.1 Service Orchestration

1.6.1.001 | Create service startup manager | 3h | P0 | G0

Acceptance: Ordered startup, dependency waiting, health verification
Dependencies: 1.4.1.013
Risks: Circular dependencies
Deliverable: internal/control/startup.go

1.6.1.002 | Implement graceful shutdown | 3h | P0 | G0

Acceptance: Signal handling, drain connections, cleanup, timeout
Dependencies: 1.6.1.001
Risks: Hanging connections
Deliverable: internal/control/shutdown.go

1.6.1.003 | Create service configuration loader | 2h | P0 | G0

Acceptance: YAML/JSON config, environment overrides, validation
Dependencies: 0.2.1.003
Risks: None
Deliverable: internal/control/config.go

1.6.1.004 | Implement leader election for control plane | 2h | P0 | G0

Acceptance: Active-standby, automatic failover, split-brain safe

Dependencies: 0.4.2.007
Risks: None
Deliverable: internal/control/leader.go

1.6.1.005 | Create service mesh between internal services | 3h | P0 | GO
Acceptance: mTLS between all services, service discovery, load balancing
Dependencies: 1.2.1.004, 1.5.1.004
Risks: Certificate management
Deliverable: internal/control/mesh.go

1.6.1.006 | Implement control plane HA | 3h | P0 | GO
Acceptance: Multiple control plane nodes, etcd quorum, shared nothing
Dependencies: 1.6.1.004
Risks: etcd performance
Deliverable: internal/control/ha.go

1.6.1.007 | Create Helm chart for control plane | 4h | P1 | OPS
Acceptance: Deployable to Kubernetes, configurable values
Dependencies: 1.6.1.001-1.6.1.006
Risks: None
Deliverable: deploy/helm/control-plane/

1.6.1.008 | Create Docker Compose production setup | 3h | P1 | OPS
Acceptance: Production-ready compose with HA, volumes, networks
Dependencies: 1.6.1.001-1.6.1.006
Risks: None
Deliverable: docker-compose.prod.yml

1.7 INTEGRATION & TESTING (Weeks 11-12)

1.7.1 Phase 1 Integration

1.7.1.001 | End-to-end node join test | 3h | P0 | QA
Acceptance: Full join flow: setup → network → register → mesh
Dependencies: 1.1.2.020, 1.2.1.015
Risks: None
Deliverable: test/e2e/node_join_test.go

1.7.1.002 | End-to-end session create test | 3h | P0 | QA
Acceptance: Create session → schedule → attach → I/O
Dependencies: 2.4.1.001, 3.1.1.001 (future phases)
Risks: Phase dependencies
Deliverable: test/e2e/session_basic_test.go

1.7.1.003 | Network partition recovery test | 3h | P0 | QA
Acceptance: Partition, detect, recover, verify consistency
Dependencies: 1.1.2.016, 1.2.1.010
Risks: None
Deliverable: test/e2e/partition_test.go

1.7.1.004 | Security end-to-end test | 3h | P0 | QA

Acceptance: mTLS, authz, attestation, rotation all verified
Dependencies: 1.5.2.005
Risks: None
Deliverable: test/e2e/security_test.go

1.7.1.005 | Performance benchmark suite | 4h | P0 | QA
Acceptance: Node join latency, gossip bandwidth, mesh throughput
Dependencies: All phase 1 tasks
Risks: None
Deliverable: test/bench/phase1_bench_test.go

1.7.1.006 | Chaos engineering - node failure | 3h | P0 | QA
Acceptance: Random node kill, cluster recovers, sessions survive
Dependencies: 1.7.1.001
Risks: None
Deliverable: test/chaos/node_failure_test.go

1.7.1.007 | Chaos engineering - network partition | 3h | P0 | QA
Acceptance: Random partitions, split-brain prevented, auto-merge
Dependencies: 1.7.1.003
Risks: None
Deliverable: test/chaos/network_partition_test.go

1.7.1.008 | Documentation: Phase 1 architecture | 4h | P0 | DOC
Acceptance: docs/architecture/phase1.md with diagrams
Dependencies: All phase 1 tasks
Risks: None
Deliverable: docs/architecture/phase1.md

1.7.1.009 | Documentation: Phase 1 operations | 4h | P0 | DOC
Acceptance: docs/operations/phase1.md with runbooks
Dependencies: 1.7.1.008
Risks: None
Deliverable: docs/operations/phase1.md

1.7.1.010 | Phase 1 retrospective and lessons learned | 2h | P0 | DOC
Acceptance: ADR-010 documenting decisions and their rationale
Dependencies: 1.7.1.008
Risks: None
Deliverable: docs/adr/010-phase1-retrospective.md

PHASE 2: RESOURCE MANAGEMENT (Weeks 13-18)

2.1 RESOURCE AGGREGATOR (Week 13)

2.1.1 Resource Collection

2.1.1.001 | Implement cgroups v2 resource reader | 3h | P0 | G0
Acceptance: CPU usage, memory, I/O per cgroup, hierarchical
Dependencies: None

Risks: cgroups v1 fallback needed
Deliverable: internal/resources/cgroups.go

2.1.1.002 | Implement /proc resource reader | 2h | P0 | G0
Acceptance: CPU info, memory info, process stats from /proc
Dependencies: None
Risks: None
Deliverable: internal/resources/proc.go

2.1.1.003 | Implement eBPF resource probe | 4h | P0 | G0
Acceptance: Kernel-level metrics without /proc overhead
Dependencies: None
Risks: eBPF program verification
Deliverable: internal/resources/ebpf.go

2.1.1.004 | Implement GPU resource collection | 3h | P0 | G0
Acceptance: Per-GPU memory, utilization, temperature from all vendors
Dependencies: 0.2.3.008
Risks: Vendor API differences
Deliverable: internal/resources/gpu.go

2.1.1.005 | Implement resource aggregation | 3h | P0 | G0
Acceptance: Sum across nodes, track available vs total
Dependencies: 2.1.1.001-2.1.1.004
Risks: Stale data
Deliverable: internal/resources/aggregate.go

2.1.1.006 | Implement resource update propagation | 2h | P0 | G0
Acceptance: Push updates to scheduler, pub/sub for changes
Dependencies: 1.3.1.002
Risks: Update frequency tuning
Deliverable: internal/resources/propagate.go

2.1.1.007 | Implement historical usage tracking | 3h | P1 | G0
Acceptance: Time-series storage, trend analysis
Dependencies: 0.4.1.011
Risks: Storage growth
Deliverable: internal/resources/history.go

2.1.1.008 | Implement utilization metrics | 2h | P0 | G0
Acceptance: CPU, memory, GPU utilization percentages
Dependencies: 2.1.1.005
Risks: None
Deliverable: internal/resources/utilization.go

2.1.1.009 | Implement capacity planning projections | 3h | P1 | G0
Acceptance: Trend-based projections, alerting thresholds
Dependencies: 2.1.1.007
Risks: Prediction accuracy
Deliverable: internal/resources/planning.go

2.1.1.010 | Implement resource quota enforcement | 2h | P0 | G0
Acceptance: Per-user quotas, hard/soft limits, enforcement
Dependencies: 0.4.1.010
Risks: None
Deliverable: internal/resources/quota.go

2.1.1.011 | Resource aggregator integration tests | 3h | P0 | QA
Acceptance: Collection, aggregation, propagation verified
Dependencies: 2.1.1.001-2.1.1.010
Risks: None
Deliverable: test/resources/integration_test.go

2.2 SCHEDULER (Weeks 13-16)

2.2.1 Omega-Model Shared State

2.2.1.001 | Implement scheduler state cache | 3h | P0 | G0
Acceptance: In-memory snapshot of cluster state, watch-based updates
Dependencies: 0.4.2.002
Risks: Memory usage
Deliverable: internal/scheduler/cache.go

2.2.1.002 | Implement optimistic concurrency control | 3h | P0 | G0
Acceptance: CAS on etcd, conflict detection, retry
Dependencies: 0.4.2.008
Risks: Contention under load
Deliverable: internal/scheduler/occ.go

2.2.1.003 | Implement scheduling queue | 3h | P0 | G0
Acceptance: Priority queue, FIFO within priority, activeQ/backoffQ
Dependencies: 0.2.1.022
Risks: None
Deliverable: internal/scheduler/queue.go

2.2.1.004 | Implement scheduling cycle | 3h | P0 | G0
Acceptance: Pop from queue, run pipeline, commit or retry
Dependencies: 2.2.1.001-2.2.1.003
Risks: Pipeline latency
Deliverable: internal/scheduler/cycle.go

2.2.1.005 | Implement scheduling parallelism | 2h | P0 | G0
Acceptance: Multiple scheduling goroutines, conflict resolution
Dependencies: 2.2.1.004
Risks: Race conditions
Deliverable: internal/scheduler/parallel.go

2.2.2 Scheduling Plugins

2.2.2.001 | Implement NodeResourcesFit plugin | 3h | P0 | G0
Acceptance: Filter by CPU/memory/GPU availability, score by fit
Dependencies: 2.1.1.005

Risks: None

Deliverable: internal/scheduler/plugins/noderesources.go

2.2.2.002 | Implement NodeAffinity plugin | 2h | P0 | G0

Acceptance: Required and preferred affinity, anti-affinity

Dependencies: 1.1.2.012

Risks: None

Deliverable: internal/scheduler/plugins/nodeaffinity.go

2.2.2.003 | Implement TopologyAware plugin | 4h | P0 | G0

Acceptance: NUMA-aware placement, PCI locality, network topology

Dependencies: 1.1.2.002

Risks: NUMA detection complexity

Deliverable: internal/scheduler/plugins/topology.go

2.2.2.004 | Implement CapabilityMatch plugin | 4h | P0 | G0

Acceptance: ClassAds expression evaluation, GPU matching, feature matching

Dependencies: 0.2.1.017

Risks: Expression evaluation correctness

Deliverable: internal/scheduler/plugins/capability.go

2.2.2.005 | Implement PrioritySort plugin | 1h | P0 | G0

Acceptance: Priority-based ordering, preemption queue

Dependencies: 2.2.1.003

Risks: None

Deliverable: internal/scheduler/plugins/priority.go

2.2.2.006 | Implement GangScheduling plugin | 4h | P0 | G0

Acceptance: All-or-nothing, group scheduling, coscheduling

Dependencies: 2.2.2.001

Risks: Head-of-line blocking

Deliverable: internal/scheduler/plugins/gang.go

2.2.2.007 | Implement LoadAware plugin | 2h | P0 | G0

Acceptance: Prefer underutilized nodes, load balancing

Dependencies: 2.1.1.008

Risks: None

Deliverable: internal/scheduler/plugins/loadaware.go

2.2.2.008 | Implement LocalityAware plugin | 3h | P1 | G0

Acceptance: Data locality scoring, cache affinity

Dependencies: 0.4.3.003

Risks: None

Deliverable: internal/scheduler/plugins/locality.go

2.2.2.009 | Implement InterPodAffinity plugin | 3h | P1 | G0

Acceptance: Co-location preferences, spread constraints

Dependencies: 2.2.2.002

Risks: $O(n^2)$ complexity

Deliverable: internal/scheduler/plugins/interpod.go

2.2.2.010 | Implement VolumeBinding plugin | 2h | P1 | G0
Acceptance: Storage locality, volume attachment
Dependencies: None
Risks: Storage complexity
Deliverable: internal/scheduler/plugins/volume.go

2.2.2.011 | Create plugin registration framework | 2h | P0 | G0
Acceptance: Dynamic plugin loading, ordering, configuration
Dependencies: 2.2.2.001-2.2.2.010
Risks: None
Deliverable: internal/scheduler/plugins/registry.go

2.2.3 Scheduling Operations

2.2.3.001 | Implement resource reservation | 3h | P0 | G0
Acceptance: Pessimistic reservation, TTL, automatic release
Dependencies: 0.4.1.007
Risks: Orphaned reservations
Deliverable: internal/scheduler/reserve.go

2.2.3.002 | Implement preemption logic | 4h | P0 | G0
Acceptance: Lower priority eviction, graceful termination
Dependencies: 2.2.2.005
Risks: Cascading preemptions
Deliverable: internal/scheduler/preempt.go

2.2.3.003 | Implement binding commit | 2h | P0 | G0
Acceptance: Atomic binding to node, resource deduction
Dependencies: 2.2.3.001
Risks: Binding failure
Deliverable: internal/scheduler/bind.go

2.2.3.004 | Implement scheduling metrics | 2h | P0 | G0
Acceptance: Scheduling latency, throughput, success rate
Dependencies: 0.2.1.010
Risks: None
Deliverable: internal/scheduler/metrics.go

2.2.3.005 | Implement scheduling events | 2h | P0 | G0
Acceptance: Scheduled, Failed, Preempted events
Dependencies: 0.3.2.003
Risks: None
Deliverable: internal/scheduler/events.go

2.2.3.006 | Implement scheduler profiling | 2h | P1 | G0
Acceptance: Per-plugin latency, decision tracing
Dependencies: 2.2.3.004
Risks: None
Deliverable: internal/scheduler/profile.go

2.2.3.007 | Create scheduler gRPC handler | 3h | P0 | G0

Acceptance: All SchedulerService RPCs implemented
Dependencies: 0.3.1.003, 2.2.1.001-2.2.3.006
Risks: None
Deliverable: internal/scheduler/server.go

2.2.3.008 | Create scheduler HTTP handler | 2h | P0 | G0

Acceptance: REST endpoints for pool, schedule, reservations
Dependencies: 2.2.3.007
Risks: None
Deliverable: internal/scheduler/http.go

2.2.3.009 | Scheduler integration tests | 4h | P0 | QA

Acceptance: All plugins, preemption, gang scheduling verified
Dependencies: 2.2.1.001-2.2.3.008
Risks: Complex test setup
Deliverable: test/scheduler/integration_test.go

2.2.3.010 | Scheduler performance benchmarks | 3h | P0 | QA

Acceptance: 1000 scheduling decisions/sec, <10ms p99 latency
Dependencies: 2.2.3.009
Risks: Hardware dependency
Deliverable: test/bench/scheduler_bench_test.go

2.3 GPU COMPUTE ENGINE (Weeks 15-16)

2.3.1 GPU Backend Implementations

2.3.1.001 | Implement NVIDIA CUDA backend | 5h | P0 | C

Acceptance: Device enum, memory alloc, kernel launch, NVML metrics
Dependencies: 0.2.3.002
Risks: Requires NVIDIA GPU
Deliverable: internal/gpu/cuda_backend.c

2.3.1.002 | Implement AMD ROCm backend | 5h | P0 | C

Acceptance: HIP device enum, memory, kernel launch, rocSMI metrics
Dependencies: 0.2.3.003
Risks: Requires AMD GPU
Deliverable: internal/gpu/rocm_backend.c

2.3.1.003 | Implement Intel oneAPI backend | 5h | P0 | C

Acceptance: Level Zero device, USM memory, SYCL kernels
Dependencies: 0.2.3.004
Risks: Requires Intel GPU
Deliverable: internal/gpu/oneapi_backend.c

2.3.1.004 | Implement Apple MLX backend | 5h | P0 | C

Acceptance: MLX device, unified memory, Metal commands
Dependencies: 0.2.3.005
Risks: Requires Apple Silicon
Deliverable: internal/gpu/mlx_backend.c

2.3.1.005 | Implement SYCL cross-platform backend | 4h | P1 | C
 Acceptance: Device selection, USM, ND-range on all vendors
 Dependencies: 0.2.3.006
 Risks: SYCL runtime availability
 Deliverable: internal/gpu/sycl_backend.c

2.3.1.006 | Create GPU backend manager | 3h | P0 | G0
 Acceptance: Auto-detect vendor, load correct backend, health checks
 Dependencies: 2.3.1.001-2.3.1.005
 Risks: None
 Deliverable: internal/gpu/manager.go

2.3.1.007 | Implement GPU memory management | 3h | P0 | G0
 Acceptance: Allocation, tracking, OOM prevention, garbage collection
 Dependencies: 2.3.1.006
 Risks: Memory leaks
 Deliverable: internal/gpu/memory.go

2.3.1.008 | Implement GPU sharing (MPS) | 3h | P0 | G0
 Acceptance: NVIDIA MPS enable/disable, fraction allocation
 Dependencies: 2.3.1.001
 Risks: MPS server management
 Deliverable: internal/gpu/mps.go

2.3.1.009 | Implement GPU time-slicing | 2h | P1 | G0
 Acceptance: Configurable slice duration, fair scheduling
 Dependencies: 2.3.1.006
 Risks: Context switch overhead
 Deliverable: internal/gpu/timeslice.go

2.3.1.010 | Implement GPU metrics collection | 2h | P0 | G0
 Acceptance: Temperature, utilization, memory, power, ECC
 Dependencies: 0.2.3.008
 Risks: None
 Deliverable: internal/gpu/metrics.go

2.3.1.011 | Create GPU compute gRPC handler | 3h | P0 | G0
 Acceptance: GPU service RPCs: execute, allocate, metrics
 Dependencies: 2.3.1.006-2.3.1.010
 Risks: None
 Deliverable: internal/gpu/server.go

2.3.1.012 | GPU compute integration tests | 4h | P0 | QA
 Acceptance: All backends, memory management, sharing tested
 Dependencies: 2.3.1.001-2.3.1.011
 Risks: GPU hardware required
 Deliverable: test/gpu/integration_test.go

2.4 HEALTH MONITOR (Weeks 17-18)

2.4.1 Monitoring Infrastructure

2.4.1.001 | Implement Prometheus metrics collector | 3h | P0 | G0
Acceptance: Custom metrics exposition, service discovery
Dependencies: 0.2.4.007
Risks: None
Deliverable: internal/health/collector.go

2.4.1.002 | Implement eBPF probe loader | 4h | P0 | G0
Acceptance: Load/unload eBPF programs, map management
Dependencies: None
Risks: Kernel version compatibility
Deliverable: internal/health/ebpf.go

2.4.1.003 | Implement node health scoring | 3h | P0 | G0
Acceptance: Composite score 0-100, weighted components
Dependencies: 2.4.1.001
Risks: Score calibration
Deliverable: internal/health/score.go

2.4.1.004 | Implement component health checks | 2h | P0 | G0
Acceptance: CPU, memory, disk, network, GPU, services
Dependencies: 2.4.1.003
Risks: None
Deliverable: internal/health/components.go

2.4.1.005 | Implement failure prediction model | 6h | P0 | ML
Acceptance: LSTM model, 85%+ accuracy, 30-90 day horizon
Dependencies: 2.4.1.001
Risks: Training data availability
Deliverable: internal/health/predictor.py

2.4.1.006 | Implement anomaly detection | 4h | P0 | ML
Acceptance: Isolation forest, real-time scoring, false positive < 5%
Dependencies: 2.4.1.001
Risks: Tuning sensitivity
Deliverable: internal/health/anomaly.py

2.4.1.007 | Implement alert generation | 2h | P0 | G0
Acceptance: Severity classification, routing, deduplication
Dependencies: 2.4.1.004, 2.4.1.006
Risks: Alert fatigue
Deliverable: internal/health/alerts.go

2.4.1.008 | Implement self-healing executor | 4h | P0 | G0
Acceptance: Memory pressure → migrate, GPU panic → evacuate, etc.
Dependencies: 2.4.1.007
Risks: Incorrect healing actions
Deliverable: internal/health/healer.go

2.4.1.009 | Create health dashboard (Grafana) | 3h | P1 | OPS
Acceptance: Node health, predictions, alerts visualized

Dependencies: 0.2.4.007
Risks: None
Deliverable: deploy/grafana/dashboards/health.json

2.4.1.010 | Create health gRPC handler | 2h | P0 | GO

Acceptance: HealthService RPCs implemented
Dependencies: 0.3.1.004, 2.4.1.001-2.4.1.008
Risks: None
Deliverable: internal/health/server.go

2.4.1.011 | Health monitor integration tests | 4h | P0 | QA

Acceptance: Metrics, prediction, anomaly, healing all verified
Dependencies: 2.4.1.001-2.4.1.010
Risks: ML model testing complexity
Deliverable: test/health/integration_test.go

2.5 PHASE 2 INTEGRATION (Week 18)

2.5.1 Integration & Testing

2.5.1.001 | End-to-end scheduling test | 3h | P0 | QA

Acceptance: Submit job → schedule → bind → execute → complete
Dependencies: 2.2.3.009, 2.3.1.012
Risks: None
Deliverable: test/e2e/scheduling_test.go

2.5.1.002 | GPU job scheduling test | 3h | P0 | QA

Acceptance: Request GPU → match capability → allocate → execute
Dependencies: 2.3.1.012
Risks: GPU hardware required
Deliverable: test/e2e/gpu_scheduling_test.go

2.5.1.003 | Preemption test | 3h | P0 | QA

Acceptance: Low priority job evicted for high priority
Dependencies: 2.2.3.002
Risks: None
Deliverable: test/e2e/preemption_test.go

2.5.1.004 | Health prediction accuracy test | 3h | P0 | QA

Acceptance: Synthetic failure, prediction triggers before failure
Dependencies: 2.4.1.005
Risks: None
Deliverable: test/e2e/prediction_test.go

2.5.1.005 | Phase 2 performance benchmarks | 4h | P0 | QA

Acceptance: 1000 jobs/sec, <10ms scheduling p99
Dependencies: All phase 2 tasks
Risks: Hardware dependency
Deliverable: test/bench/phase2_bench_test.go

2.5.1.006 | Documentation: Phase 2 architecture | 4h | P0 | DOC

Acceptance: docs/architecture/phase2.md with diagrams
Dependencies: All phase 2 tasks
Risks: None
Deliverable: docs/architecture/phase2.md

2.5.1.007 | Documentation: Phase 2 operations | 4h | P0 | DOC
Acceptance: docs/operations/phase2.md with GPU management runbooks
Dependencies: 2.5.1.006
Risks: None
Deliverable: docs/operations/phase2.md

2.5.1.008 | Phase 2 retrospective | 2h | P0 | DOC
Acceptance: ADR-020 documenting decisions
Dependencies: 2.5.1.006
Risks: None
Deliverable: docs/adr/020-phase2-retrospective.md

PHASE 3: SESSION MANAGER (Weeks 19-26)

3.1 SESSION BACKEND ABSTRACTION (Weeks 19-20)

3.1.1 Backend Interface

3.1.1.001 | Define SessionBackend interface | 2h | P0 | G0
Acceptance: All lifecycle, I/O, migration methods defined
Dependencies: 0.3.1.002
Risks: None
Deliverable: internal/session/backend.go

3.1.1.002 | Implement backend factory | 2h | P0 | G0
Acceptance: Create backend by type (tmux, zellij, screen, native)
Dependencies: 3.1.1.001
Risks: None
Deliverable: internal/session/factory.go

3.1.1.003 | Implement backend health checker | 2h | P0 | G0
Acceptance: Periodic health checks, unhealthy backend detection
Dependencies: 3.1.1.002
Risks: None
Deliverable: internal/session/backend_health.go

3.1.1.004 | Implement backend metrics | 1h | P1 | G0
Acceptance: Backend creation, operation counts, latency
Dependencies: 0.2.1.010
Risks: None
Deliverable: internal/session/backend_metrics.go

3.1.2 tmux Backend

3.1.2.001 | Implement tmux process management | 3h | P0 | G0

Acceptance: Start/stop tmux server, socket management
Dependencies: None
Risks: tmux version compatibility
Deliverable: internal/session/tmux/process.go

3.1.2.002 | Implement tmux control mode client | 4h | P0 | G0

Acceptance: Connect via -C, send commands, parse responses
Dependencies: 3.1.2.001
Risks: Control mode protocol complexity
Deliverable: internal/session/tmux/control.go

3.1.2.003 | Implement tmux session CRUD | 3h | P0 | G0

Acceptance: Create, list, rename, kill sessions via control mode
Dependencies: 3.1.2.002
Risks: None
Deliverable: internal/session/tmux/sessions.go

3.1.2.004 | Implement tmux window management | 3h | P0 | G0

Acceptance: Create, rename, select, kill windows
Dependencies: 3.1.2.003
Risks: None
Deliverable: internal/session/tmux/windows.go

3.1.2.005 | Implement tmux pane management | 3h | P0 | G0

Acceptance: Split, resize, zoom, kill panes
Dependencies: 3.1.2.004
Risks: None
Deliverable: internal/session/tmux/panes.go

3.1.2.006 | Implement tmux I/O capture | 4h | P0 | G0

Acceptance: Capture pane output via %output events
Dependencies: 3.1.2.005
Risks: Event parsing complexity
Deliverable: internal/session/tmux/output.go

3.1.2.007 | Implement tmux input injection | 3h | P0 | G0

Acceptance: Send keys to pane, special key handling
Dependencies: 3.1.2.006
Risks: None
Deliverable: internal/session/tmux/input.go

3.1.2.008 | Implement tmux resize handling | 2h | P0 | G0

Acceptance: Resize pane, window, client dimensions
Dependencies: 3.1.2.005
Risks: None
Deliverable: internal/session/tmux/resize.go

3.1.2.009 | Implement tmux layout management | 3h | P1 | G0

Acceptance: Parse and apply layouts, custom layouts
Dependencies: 3.1.2.005
Risks: Layout string format

Deliverable: internal/session/tmux/layout.go

3.1.2.010 | Implement tmux notification handling | 2h | P0 | GO

Acceptance: Subscribe to %session-changed, %window-add, etc.

Dependencies: 3.1.2.002

Risks: None

Deliverable: internal/session/tmux/notify.go

3.1.2.011 | Implement tmux environment management | 2h | P0 | GO

Acceptance: Set/get environment variables, update command

Dependencies: 3.1.2.003

Risks: None

Deliverable: internal/session/tmux/env.go

3.1.2.012 | tmux backend integration tests | 4h | P0 | QA

Acceptance: Full CRUD, I/O, resize, notification cycle

Dependencies: 3.1.2.001-3.1.2.011

Risks: tmux must be installed

Deliverable: test/session/tmux_test.go

3.1.3 Zellij Backend

3.1.3.001 | Implement Zellij IPC client | 3h | P0 | GO

Acceptance: Connect to Zellij server, send actions, receive events

Dependencies: None

Risks: Zellij IPC protocol changes

Deliverable: internal/session/zellij/ipc.go

3.1.3.002 | Implement Zellij session CRUD | 3h | P0 | GO

Acceptance: Create, list, attach, delete sessions

Dependencies: 3.1.3.001

Risks: None

Deliverable: internal/session/zellij/sessions.go

3.1.3.003 | Implement Zellij tab/pane management | 3h | P0 | GO

Acceptance: Create, rename, move, resize tabs and panes

Dependencies: 3.1.3.002

Risks: None

Deliverable: internal/session/zellij/panes.go

3.1.3.004 | Implement Zellij I/O handling | 3h | P0 | GO

Acceptance: PTY I/O forwarding through Zellij

Dependencies: 3.1.3.003

Risks: None

Deliverable: internal/session/zellij/io.go

3.1.3.005 | Implement Zellij plugin API | 3h | P1 | GO

Acceptance: Load custom plugins, WebAssembly execution

Dependencies: 3.1.3.001

Risks: WASI complexity

Deliverable: internal/session/zellij/plugins.go

3.1.3.006 | Zellij backend integration tests | 3h | P0 | QA

Acceptance: Full session lifecycle, I/O verified

Dependencies: 3.1.3.001-3.1.3.005

Risks: Zellij must be installed

Deliverable: test/session/zellij_test.go

3.1.4 screen Backend

3.1.4.001 | Implement screen process management | 3h | P0 | GO

Acceptance: Start/stop screen, socket management

Dependencies: None

Risks: screen version compatibility

Deliverable: internal/session/screen/process.go

3.1.4.002 | Implement screen session CRUD | 3h | P0 | GO

Acceptance: Create, list, attach, kill sessions

Dependencies: 3.1.4.001

Risks: None

Deliverable: internal/session/screen/sessions.go

3.1.4.003 | Implement screen window management | 2h | P0 | GO

Acceptance: Create, switch, rename windows

Dependencies: 3.1.4.002

Risks: None

Deliverable: internal/session/screen/windows.go

3.1.4.004 | Implement screen I/O handling | 2h | P0 | GO

Acceptance: PTY I/O through screen

Dependencies: 3.1.4.003

Risks: None

Deliverable: internal/session/screen/io.go

3.1.4.005 | screen backend integration tests | 2h | P0 | QA

Acceptance: Full session lifecycle verified

Dependencies: 3.1.4.001-3.1.4.004

Risks: screen must be installed

Deliverable: test/session/screen_test.go

3.1.5 Native Backend

3.1.5.001 | Implement native PTY backend | 4h | P0 | GO

Acceptance: Direct PTY without multiplexer, shell execution

Dependencies: 0.2.2.009

Risks: Platform differences

Deliverable: internal/session/native/pty.go

3.1.5.002 | Implement native process management | 3h | P0 | GO

Acceptance: Start, signal, wait for processes

Dependencies: 3.1.5.001

Risks: None

Deliverable: internal/session/native/process.go

3.1.5.003 | Implement native I/O handling | 2h | P0 | G0

Acceptance: Direct PTY I/O without intermediate

Dependencies: 3.1.5.002

Risks: None

Deliverable: internal/session/native/io.go

3.1.5.004 | Native backend integration tests | 2h | P0 | QA

Acceptance: Process lifecycle, I/O verified

Dependencies: 3.1.5.001-3.1.5.003

Risks: None

Deliverable: test/session/native_test.go

3.2 DISTRIBUTED SESSION CORE (Weeks 20-22)

3.2.1 Session Lifecycle

3.2.1.001 | Implement session creation | 3h | P0 | G0

Acceptance: Create session, allocate resources, start backend

Dependencies: 2.2.3.003, 3.1.1.002

Risks: Resource allocation failure

Deliverable: internal/session/create.go

3.2.1.002 | Implement session attachment | 3h | P0 | G0

Acceptance: Attach client, establish I/O stream, resume state

Dependencies: 3.2.1.001, 0.2.1.016

Risks: Concurrent attachments

Deliverable: internal/session/attach.go

3.2.1.003 | Implement session detachment | 2h | P0 | G0

Acceptance: Client disconnects, session continues, cleanup

Dependencies: 3.2.1.002

Risks: None

Deliverable: internal/session/detach.go

3.2.1.004 | Implement session termination | 2h | P0 | G0

Acceptance: Graceful shutdown, resource release, event publish

Dependencies: 3.2.1.001

Risks: Orphaned resources

Deliverable: internal/session/terminate.go

3.2.1.005 | Implement session listing | 2h | P0 | G0

Acceptance: List by owner, status, node; pagination

Dependencies: 0.4.2.010

Risks: None

Deliverable: internal/session/list.go

3.2.1.006 | Implement session state machine | 3h | P0 | G0

Acceptance: CREATING-RUNNING-MIGRATING-PAUSED-TERMINATED

Dependencies: 3.2.1.001-3.2.1.004

Risks: Invalid transitions
Deliverable: internal/session/statemachine.go

3.2.2 Window Management

3.2.2.001 | Implement window CRUD | 2h | P0 | G0
Acceptance: Create, rename, reorder, delete windows
Dependencies: 3.1.2.004, 3.1.3.003
Risks: None
Deliverable: internal/session/window_crud.go

3.2.2.002 | Implement window layout management | 3h | P0 | G0
Acceptance: Tiled, floating, custom layouts
Dependencies: 3.1.2.009
Risks: None
Deliverable: internal/session/window_layout.go

3.2.2.003 | Implement window focus tracking | 2h | P0 | G0
Acceptance: Active window, focus history, notification
Dependencies: 3.2.2.001
Risks: None
Deliverable: internal/session/window_focus.go

3.2.2.004 | Implement window CRDT sync | 4h | P0 | G0
Acceptance: Yjs-style CRDT for distributed window state
Dependencies: 3.2.2.001
Risks: CRDT merge correctness
Deliverable: internal/session/window_crdt.go

3.2.3 Distributed Pane Placement

3.2.3.001 | Implement pane scheduling | 3h | P0 | G0
Acceptance: Schedule pane to best node, GPU if needed
Dependencies: 2.2.2.004
Risks: Cross-node I/O latency
Deliverable: internal/session/pane_schedule.go

3.2.3.002 | Implement pane I/O proxy | 4h | P0 | G0
Acceptance: Proxy PTY I/O across nodes via ZeroMQ
Dependencies: 0.2.2.004, 1.3.1.004
Risks: Latency for remote panes
Deliverable: internal/session/pane_proxy.go

3.2.3.003 | Implement pane CRDT sync | 4h | P0 | G0
Acceptance: Synchronize pane state across nodes
Dependencies: 3.2.3.002
Risks: CRDT complexity
Deliverable: internal/session/pane_crdt.go

3.2.3.004 | Implement pane GPU attachment | 3h | P0 | G0
Acceptance: Attach GPU to pane, schedule GPU job

Dependencies: 2.3.1.006
Risks: GPU availability
Deliverable: internal/session/pane_gpu.go

3.3 I/O FORWARDING (Weeks 22-24)

3.3.1 PTY Management

3.3.1.001 | Implement PTY allocation (Linux) | 2h | P0 | GO
Acceptance: Open PTY master/slave, set termios
Dependencies: None
Risks: None
Deliverable: internal/pty/pty_linux.go

3.3.1.002 | Implement PTY allocation (macOS) | 2h | P0 | GO
Acceptance: Open PTY on macOS using posix_openpt
Dependencies: 3.3.1.001
Risks: Platform differences
Deliverable: internal/pty/pty_darwin.go

3.3.1.003 | Implement PTY resize | 1h | P0 | GO
Acceptance: TIOCSWINSZ ioctl for row/col changes
Dependencies: 3.3.1.001
Risks: None
Deliverable: internal/pty/resize.go

3.3.1.004 | Implement PTY signal forwarding | 1h | P0 | GO
Acceptance: Forward SIGINT, SIGTERM, SIGWINCH
Dependencies: 3.3.1.001
Risks: None
Deliverable: internal/pty/signal.go

3.3.1.005 | Implement PTY I/O multiplexer | 3h | P0 | GO
Acceptance: Read/write PTY with select/epoll
Dependencies: 3.3.1.001
Risks: Blocking I/O
Deliverable: internal/pty/mux.go

3.3.2 WebSocket I/O Stream

3.3.2.001 | Implement WebSocket server for I/O | 3h | P0 | GO
Acceptance: Upgrade HTTP to WebSocket, handle subprotocols
Dependencies: 0.2.1.016, 1.4.1.012
Risks: Connection limits
Deliverable: internal/io/websocket.go

3.3.2.002 | Implement I/O message framing | 2h | P0 | GO
Acceptance: Binary framing for output, text for commands
Dependencies: 3.3.2.001
Risks: None
Deliverable: internal/io/frame.go

3.3.2.003 | Implement I/O compression | 2h | P1 | G0
Acceptance: Per-message deflate, reduce bandwidth 60%+
Dependencies: 3.3.2.002
Risks: CPU overhead
Deliverable: internal/io/compress.go

3.3.2.004 | Implement I/O encryption | 2h | P0 | G0
Acceptance: Encrypt I/O stream over WebSocket (already mTLS)
Dependencies: 3.3.2.001
Risks: None
Deliverable: internal/io/encrypt.go

3.3.2.005 | Implement I/O heartbeat | 1h | P0 | G0
Acceptance: Ping/pong, detect stale connections
Dependencies: 3.3.2.001
Risks: None
Deliverable: internal/io/heartbeat.go

3.3.2.006 | Implement I/O reconnection | 3h | P0 | G0
Acceptance: Client reconnects, resume from last position
Dependencies: 3.3.2.005
Risks: State synchronization
Deliverable: internal/io/reconnect.go

3.3.2.007 | Implement input handling | 2h | P0 | G0
Acceptance: Keyboard input, special keys, mouse events
Dependencies: 3.3.2.002
Risks: Key encoding differences
Deliverable: internal/io/input.go

3.3.2.008 | Implement output rendering | 2h | P0 | G0
Acceptance: VT100, xterm, 256color support
Dependencies: 3.3.2.002
Risks: Terminal emulation complexity
Deliverable: internal/io/output.go

3.3.2.009 | Implement I/O buffering | 2h | P0 | G0
Acceptance: Ring buffer for output, flow control
Dependencies: 3.3.2.008
Risks: Memory usage
Deliverable: internal/io/buffer.go

3.3.2.010 | WebSocket I/O integration tests | 4h | P0 | QA
Acceptance: Full I/O cycle, reconnection, compression
Dependencies: 3.3.2.001-3.3.2.009
Risks: None
Deliverable: test/io/websocket_test.go

3.4 SESSION MIGRATION (Weeks 24-26)

3.4.1 CRIU Integration

3.4.1.001 | Implement CRIU checkpoint wrapper | 4h | P0 | G0
Acceptance: Shell out to criu dump, capture all options
Dependencies: None
Risks: CRIU version compatibility
Deliverable: internal/migration/criu/checkpoint.go

3.4.1.002 | Implement CRIU restore wrapper | 4h | P0 | G0
Acceptance: Shell out to criu restore, handle failures
Dependencies: 3.4.1.001
Risks: Restore failures
Deliverable: internal/migration/criu/restore.go

3.4.1.003 | Implement TCP repair handling | 3h | P0 | G0
Acceptance: TCP_REPAIR mode, connection state capture
Dependencies: 3.4.1.001
Risks: Kernel support required
Deliverable: internal/migration/criu/tcp.go

3.4.1.004 | Implement PTY state capture | 3h | P0 | G0
Acceptance: Terminal state, scrollbar buffer
Dependencies: 3.4.1.001
Risks: Large state files
Deliverable: internal/migration/criu/pty.go

3.4.1.005 | Implement checkpoint streaming | 3h | P0 | G0
Acceptance: Stream checkpoint files via Arrow Flight
Dependencies: 3.4.1.001
Risks: Bandwidth usage
Deliverable: internal/migration/criu/stream.go

3.4.1.006 | CRIU integration tests | 3h | P0 | QA
Acceptance: Checkpoint and restore simple process
Dependencies: 3.4.1.001-3.4.1.005
Risks: Requires root and CRIU
Deliverable: test/migration/criu_test.go

3.4.2 DMTCP Integration

3.4.2.001 | Implement DMTCP checkpoint wrapper | 3h | P0 | G0
Acceptance: Shell out to dmtcp_checkpoint
Dependencies: None
Risks: DMTCP version compatibility
Deliverable: internal/migration/dmtcp/checkpoint.go

3.4.2.002 | Implement DMTCP restore wrapper | 3h | P0 | G0
Acceptance: Shell out to dmtcp_restart
Dependencies: 3.4.2.001
Risks: Restore environment differences

Deliverable: internal/migration/dmtcp/restore.go

3.4.2.003 | DMTCP integration tests | 2h | P0 | QA

Acceptance: Checkpoint and restore via DMTCP

Dependencies: 3.4.2.001, 3.4.2.002

Risks: Requires DMTCP

Deliverable: test/migration/dmtcp_test.go

3.4.3 Migration Orchestration

3.4.3.001 | Implement migration decision engine | 3h | P0 | G0

Acceptance: Trigger migration on node failure, resource pressure

Dependencies: 2.4.1.008

Risks: Incorrect triggers

Deliverable: internal/migration/decide.go

3.4.3.002 | Implement migration coordinator | 4h | P0 | G0

Acceptance: Coordinate checkpoint → transfer → restore → handover

Dependencies: 3.4.1.001-3.4.2.003

Risks: Race conditions

Deliverable: internal/migration/coordinate.go

3.4.3.003 | Implement client handover | 3h | P0 | G0

Acceptance: Redirect client I/O to new node seamlessly

Dependencies: 3.3.2.006

Risks: Brief disconnection

Deliverable: internal/migration/handover.go

3.4.3.004 | Implement migration rollback | 2h | P0 | G0

Acceptance: If migration fails, continue on original node

Dependencies: 3.4.3.002

Risks: Data inconsistency

Deliverable: internal/migration/rollback.go

3.4.3.005 | Implement migration metrics | 1h | P1 | G0

Acceptance: Duration, data size, success rate

Dependencies: 0.2.1.010

Risks: None

Deliverable: internal/migration/metrics.go

3.4.3.006 | Migration integration tests | 4h | P0 | QA

Acceptance: Full migration cycle, rollback, failure scenarios

Dependencies: 3.4.3.001-3.4.3.005

Risks: Complex test setup

Deliverable: test/migration/integration_test.go

3.5 HTMUX CLI (Weeks 25-26)

3.5.1 CLI Framework

3.5.1.001 | Set up cobra CLI framework | 2h | P0 | G0

Acceptance: Command structure, help generation
Dependencies: None
Risks: None
Deliverable: cmd/htmux/main.go

3.5.1.002 | Implement config command | 2h | P0 | G0
Acceptance: Read ~/.htmux/config.yaml, show/set values
Dependencies: 0.2.1.003
Risks: None
Deliverable: cmd/htmux/config.go

3.5.1.003 | Implement server connection | 2h | P0 | G0
Acceptance: Connect to control plane, mTLS auth, SPIFFE
Dependencies: 1.5.1.004
Risks: Certificate paths
Deliverable: cmd/htmux/client.go

3.5.1.004 | Implement session list command | 1h | P0 | G0
Acceptance: htmux ls shows all sessions with status
Dependencies: 3.5.1.003
Risks: None
Deliverable: cmd/htmux/list.go

3.5.1.005 | Implement session create command | 2h | P0 | G0
Acceptance: htmux new -s name --mode=batch --gpu=nvidia
Dependencies: 3.5.1.003
Risks: None
Deliverable: cmd/htmux/new.go

3.5.1.006 | Implement session attach command | 3h | P0 | G0
Acceptance: htmux attach -t name, WebSocket I/O
Dependencies: 3.5.1.005, 3.3.2.001
Risks: Terminal raw mode
Deliverable: cmd/htmux/attach.go

3.5.1.007 | Implement session kill command | 1h | P0 | G0
Acceptance: htmux kill-session -t name
Dependencies: 3.5.1.003
Risks: None
Deliverable: cmd/htmux/kill.go

3.5.1.008 | Implement window commands | 2h | P0 | G0
Acceptance: htmux new-window, select-window, kill-window
Dependencies: 3.5.1.006
Risks: None
Deliverable: cmd/htmux/window.go

3.5.1.009 | Implement pane commands | 2h | P0 | G0
Acceptance: htmux split-window, resize-pane, send-keys
Dependencies: 3.5.1.008
Risks: None

Deliverable: cmd/htmux/pane.go

3.5.1.010 | Implement cluster status command | 2h | P0 | GO

Acceptance: htmux status shows nodes, resources, sessions

Dependencies: 3.5.1.003

Risks: None

Deliverable: cmd/htmux/status.go

3.5.1.011 | Implement node management commands | 2h | P0 | GO

Acceptance: htmux node list, node show, node remove

Dependencies: 3.5.1.003

Risks: None

Deliverable: cmd/htmux/node.go

3.5.1.012 | Implement GPU info command | 1h | P1 | GO

Acceptance: htmux gpu shows cluster GPU inventory

Dependencies: 3.5.1.003

Risks: None

Deliverable: cmd/htmux/gpu.go

3.5.1.013 | Implement shell completions | 2h | P1 | GO

Acceptance: bash, zsh, fish completions

Dependencies: 3.5.1.001

Risks: None

Deliverable: cmd/htmux/completion.go

3.5.1.014 | Implement man page generation | 2h | P1 | GO

Acceptance: man htmux shows all commands

Dependencies: 3.5.1.001

Risks: None

Deliverable: cmd/htmux/man.go

3.5.1.015 | CLI integration tests | 4h | P0 | QA

Acceptance: All commands tested against mock server

Dependencies: 3.5.1.001-3.5.1.014

Risks: None

Deliverable: test/cli/integration_test.go

3.6 PHASE 3 INTEGRATION (Week 26)

3.6.1.001 | End-to-end session lifecycle test | 3h | P0 | QA

Acceptance: Create → attach → I/O → detach → reattach → kill

Dependencies: 3.5.1.015

Risks: None

Deliverable: test/e2e/session_lifecycle_test.go

3.6.1.002 | End-to-end migration test | 3h | P0 | QA

Acceptance: Session migrates while active, client reconnects

Dependencies: 3.4.3.006

Risks: None

Deliverable: test/e2e/session_migration_test.go

3.6.1.003 | End-to-end distributed pane test | 3h | P0 | QA
Acceptance: Panes on different nodes, I/O works
Dependencies: 3.2.3.004
Risks: None
Deliverable: test/e2e/distributed_pane_test.go

3.6.1.004 | Phase 3 chaos tests | 3h | P0 | QA
Acceptance: Kill node during session, auto-migration
Dependencies: 3.6.1.002
Risks: None
Deliverable: test/chaos/session_failure_test.go

3.6.1.005 | Documentation: Session architecture | 4h | P0 | DOC
Acceptance: docs/architecture/session.md with diagrams
Dependencies: All phase 3 tasks
Risks: None
Deliverable: docs/architecture/session.md

3.6.1.006 | Documentation: User guide | 4h | P0 | DOC
Acceptance: docs/user-guide.md with examples
Dependencies: 3.6.1.005
Risks: None
Deliverable: docs/user-guide.md

3.6.1.007 | Phase 3 retrospective | 2h | P0 | DOC
Acceptance: ADR-030 documenting decisions
Dependencies: 3.6.1.005
Risks: None
Deliverable: docs/adr/030-phase3-retrospective.md

PHASE 4: BUILD SERVICE (Weeks 27-30)

4.1 BAZEL RBE IMPLEMENTATION (Weeks 27-28)

4.1.1.001 | Implement RBE protocol server | 5h | P0 | G0
Acceptance: Execute, WaitExecution, GetCapabilities RPCs
Dependencies: 0.3.1.007
Risks: Protocol complexity
Deliverable: internal/build/rbe/server.go

4.1.1.002 | Implement action cache | 4h | P0 | G0
Acceptance: Content-addressed storage, GetActionResult, UpdateActionResult
Dependencies: 0.4.3.001
Risks: Cache invalidation
Deliverable: internal/build/rbe/action_cache.go

4.1.1.003 | Implement CAS (Content-Addressed Storage) | 4h | P0 | G0
Acceptance: FindMissingBlobs, BatchUpdateBlobs, BatchReadBlobs, GetTree

Dependencies: 4.1.1.002
Risks: Storage growth
Deliverable: internal/build/rbe/cas.go

4.1.1.004 | Implement execution service | 5h | P0 | GO
Acceptance: Execute RPC, streaming responses, job queuing
Dependencies: 4.1.1.001
Risks: Execution isolation
Deliverable: internal/build/rbe/execution.go

4.1.1.005 | Implement worker pool management | 4h | P0 | GO
Acceptance: Worker registration, lease management, health checks
Dependencies: 4.1.1.004
Risks: Worker failures
Deliverable: internal/build/rbe/workers.go

4.1.1.006 | Implement Buildbarn-compatible API | 3h | P0 | GO
Acceptance: Compatible with bb-scheduler, bb-storage
Dependencies: 4.1.1.001-4.1.1.005
Risks: Version compatibility
Deliverable: internal/build/rbe/buildbarn.go

4.1.1.007 | RBE server integration tests | 4h | P0 | QA
Acceptance: Full RBE flow: upload → execute → download results
Dependencies: 4.1.1.001-4.1.1.006
Risks: None
Deliverable: test/build/rbe_test.go

4.2 AOSP INTEGRATION (Weeks 29-30)

4.2.1.001 | Implement AOSP build detection | 2h | P0 | GO
Acceptance: Detect Android.bp, Android.mk files
Dependencies: None
Risks: None
Deliverable: internal/build/aosp/detect.go

4.2.1.002 | Implement Soong/Blueprint analysis | 3h | P0 | GO
Acceptance: Parse Android.bp, extract module dependencies
Dependencies: 4.2.1.001
Risks: Blueprint format changes
Deliverable: internal/build/aosp/soong.go

4.2.1.003 | Implement distcc worker pool | 4h | P0 | GO
Acceptance: Register workers, distribute compilation jobs
Dependencies: 4.1.1.005
Risks: None
Deliverable: internal/build/aosp/distcc.go

4.2.1.004 | Implement ccache/sccache integration | 3h | P0 | GO
Acceptance: Shared cache across nodes, hit rate tracking
Dependencies: 4.2.1.003

Risks: Cache coherence
Deliverable: internal/build/aosp/cache.go

4.2.1.005 | Implement Ninja job distribution | 3h | P0 | GO

Acceptance: Distribute ninja jobs, aggregate results

Dependencies: 4.2.1.003

Risks: Job dependencies

Deliverable: internal/build/aosp/ninja.go

4.2.1.006 | Implement build progress reporting | 2h | P0 | GO

Acceptance: Real-time progress, ETA, failure notification

Dependencies: 4.1.1.004

Risks: None

Deliverable: internal/build/aosp/progress.go

4.2.1.007 | AOSP build integration test | 4h | P0 | QA

Acceptance: Build hello-world Android module distributed

Dependencies: 4.2.1.001-4.2.1.006

Risks: Requires AOSP checkout

Deliverable: test/build/aosp_test.go

4.3 PHASE 4 INTEGRATION

4.3.1.001 | End-to-end AOSP build test | 4h | P0 | QA

Acceptance: Full AOSP build distributed across 3 nodes

Dependencies: 4.2.1.007

Risks: Hours-long build

Deliverable: test/e2e/aosp_build_test.go

4.3.1.002 | Documentation: Build service | 3h | P0 | DOC

Acceptance: docs/build-service.md with setup instructions

Dependencies: All phase 4 tasks

Risks: None

Deliverable: docs/build-service.md

4.3.1.003 | Phase 4 retrospective | 2h | P0 | DOC

Acceptance: ADR-040 documenting decisions

Dependencies: 4.3.1.002

Risks: None

Deliverable: docs/adr/040-phase4-retrospective.md

PHASE 5: LLM BRAIN (Weeks 31-36)

5.1 LLMSVERIFIER INTEGRATION (Week 31)

5.1.1.001 | Create LLMSVerifier Go SDK wrapper | 3h | P0 | GO

Acceptance: Initialize client, configure providers

Dependencies: None

Risks: API compatibility

Deliverable: internal/llm/verifier.go

5.1.1.002 | Implement provider adapter for Kimi | 2h | P0 | G0

Acceptance: Send requests, parse responses, handle errors

Dependencies: 5.1.1.001

Risks: API changes

Deliverable: internal/llm/providers/kimi.go

5.1.1.003 | Implement provider adapter for DeepSeek | 2h | P0 | G0

Acceptance: DeepSeek V4 API integration

Dependencies: 5.1.1.001

Risks: None

Deliverable: internal/llm/providers/deepseek.go

5.1.1.004 | Implement provider adapter for Claude | 2h | P0 | G0

Acceptance: Anthropic API integration

Dependencies: 5.1.1.001

Risks: None

Deliverable: internal/llm/providers/claude.go

5.1.1.005 | Implement circuit breaker | 2h | P0 | G0

Acceptance: Fail fast on provider outage, automatic recovery

Dependencies: 5.1.1.001

Risks: None

Deliverable: internal/llm/circuit.go

5.1.1.006 | Implement request timeout and retry | 2h | P0 | G0

Acceptance: Configurable timeout, exponential backoff retry

Dependencies: 5.1.1.005

Risks: None

Deliverable: internal/llm/retry.go

5.1.1.007 | Implement response caching | 2h | P1 | G0

Acceptance: Cache similar prompts, TTL-based invalidation

Dependencies: 0.4.3.001

Risks: Stale responses

Deliverable: internal/llm/cache.go

5.1.1.008 | LLMsVerifier integration tests | 3h | P0 | QA

Acceptance: All providers, circuit breaker, retry, cache

Dependencies: 5.1.1.001-5.1.1.007

Risks: API keys required

Deliverable: test/llm/verifier_test.go

5.2 ADVISORY SYSTEM (Weeks 32-34)

5.2.1.001 | Implement RAG knowledge base | 5h | P0 | G0

Acceptance: Document ingestion, vector storage, similarity search

Dependencies: None

Risks: Embedding model selection

Deliverable: internal/llm/rag.go

5.2.1.002 | Implement context window management | 3h | P0 | G0

Acceptance: Token counting, truncation, summarization

Dependencies: 5.2.1.001

Risks: Token counting accuracy

Deliverable: internal/llm/context.go

5.2.1.003 | Implement chain-of-thought generation | 4h | P0 | G0

Acceptance: Step-by-step reasoning, tool calls

Dependencies: 5.1.1.001

Risks: Reasoning quality

Deliverable: internal/llm/cot.go

5.2.1.004 | Implement advisory creation pipeline | 4h | P0 | G0

Acceptance: Input: metrics+events → Output: structured advisory

Dependencies: 5.2.1.003

Risks: Advisory quality

Deliverable: internal/llm/pipeline.go

5.2.1.005 | Implement risk assessment scoring | 3h | P0 | G0

Acceptance: Risk matrix: impact × probability, classification

Dependencies: 5.2.1.004

Risks: None

Deliverable: internal/llm/risk.go

5.2.1.006 | Implement auto-approval logic | 3h | P0 | G0

Acceptance: Low risk + high confidence → auto-approve

Dependencies: 5.2.1.005

Risks: Incorrect auto-approvals

Deliverable: internal/llm/autoapprove.go

5.2.1.007 | Implement human review queue | 3h | P0 | G0

Acceptance: Queue medium/high risk advisories, notify operators

Dependencies: 5.2.1.006

Risks: Notification delivery

Deliverable: internal/llm/review.go

5.2.1.008 | Create advisory gRPC handler | 2h | P0 | G0

Acceptance: AdvisoryService RPCs implemented

Dependencies: 0.3.1.005, 5.2.1.004-5.2.1.007

Risks: None

Deliverable: internal/llm/server.go

5.2.1.009 | Advisory system integration tests | 4h | P0 | QA

Acceptance: Full pipeline: event → advisory → approval

Dependencies: 5.2.1.001-5.2.1.008

Risks: LLM API dependency

Deliverable: test/llm/advisory_test.go

5.3 LEARNING & ADAPTATION (Week 35)

5.3.1.001 | Implement metrics ingestion pipeline | 3h | P0 | ML
Acceptance: Pull from Prometheus, normalize, store
Dependencies: 2.4.1.001
Risks: None
Deliverable: internal/llm/learning/ingest.py

5.3.1.002 | Implement pattern recognition | 4h | P0 | ML
Acceptance: Detect recurring patterns, correlate events
Dependencies: 5.3.1.001
Risks: False positives
Deliverable: internal/llm/learning/patterns.py

5.3.1.003 | Implement RL feedback loop | 4h | P1 | ML
Acceptance: Learn from approved/rejected advisories
Dependencies: 5.3.1.002
Risks: Slow convergence
Deliverable: internal/llm/learning/rl.py

5.3.1.004 | Implement configuration optimization | 3h | P1 | ML
Acceptance: Suggest config changes, predict impact
Dependencies: 5.3.1.002
Risks: Incorrect suggestions
Deliverable: internal/llm/learning/optimize.py

5.3.1.005 | Learning system tests | 3h | P0 | QA
Acceptance: Pattern recognition, feedback loop verified
Dependencies: 5.3.1.001-5.3.1.004
Risks: None
Deliverable: test/llm/learning_test.py

5.4 CONSTITUTIONAL ENFORCEMENT (Week 36)

5.4.1.001 | Implement HelixConstitution parser | 3h | P0 | GO
Acceptance: Parse constitution markdown, extract rules
Dependencies: None
Risks: Format changes
Deliverable: internal/llm/constitution/parser.go

5.4.1.002 | Implement safety constraint validator | 3h | P0 | GO
Acceptance: Check advisory against constraints, reject if violates
Dependencies: 5.4.1.001
Risks: Constraint interpretation
Deliverable: internal/llm/constitution/validate.go

5.4.1.003 | Implement explanation generator | 2h | P0 | GO
Acceptance: Why advisory was approved/rejected, full trace
Dependencies: 5.4.1.002
Risks: None
Deliverable: internal/llm/constitution/explain.go

5.4.1.004 | Constitutional enforcement tests | 3h | P0 | QA
Acceptance: Rules parsed, violations caught, explanations generated
Dependencies: 5.4.1.001-5.4.1.003
Risks: None
Deliverable: test/llm/constitution_test.go

5.4.1.005 | Documentation: LLM Brain | 4h | P0 | DOC
Acceptance: docs/architecture/llm-brain.md
Dependencies: All phase 5 tasks
Risks: None
Deliverable: docs/architecture/llm-brain.md

5.4.1.006 | Phase 5 retrospective | 2h | P0 | DOC
Acceptance: ADR-050 documenting decisions
Dependencies: 5.4.1.005
Risks: None
Deliverable: docs/adr/050-phase5-retrospective.md

PHASE 6: SECURITY HARDENING (Weeks 37-40)

6.1 ZERO TRUST IMPLEMENTATION

6.1.1.001 | Implement certificate rotation automation | 3h | P0 | SEC
Acceptance: Automatic rotation before expiry, no downtime
Dependencies: 1.5.1.003
Risks: Clock skew
Deliverable: internal/security/rotation.go

6.1.1.002 | Implement secrets management (Vault) | 3h | P0 | SEC
Acceptance: Dynamic secrets, automatic revocation
Dependencies: 0.2.4.009
Risks: Vault availability
Deliverable: internal/security/vault.go

6.1.1.003 | Implement network policies | 3h | P0 | SEC
Acceptance: Micro-segmentation, ingress/egress rules
Dependencies: 1.2.1.004
Risks: Overly restrictive
Deliverable: internal/security/netpol.go

6.1.1.004 | Implement runtime security (seccomp) | 3h | P0 | SEC
Acceptance: System call filtering, AppArmor profiles
Dependencies: None
Risks: Breaking legitimate calls
Deliverable: internal/security/seccomp.go

6.1.1.005 | Implement supply chain security | 3h | P1 | SEC
Acceptance: SBOM generation, SLSA compliance, sigstore
Dependencies: None

Risks: Tool availability
Deliverable: internal/security/supply.go

6.1.1.006 | Security audit and penetration test | 8h | P0 | SEC
Acceptance: Third-party security assessment, all findings addressed
Dependencies: 6.1.1.001-6.1.1.005
Risks: Findings requiring architecture changes
Deliverable: docs/security/audit-report.pdf

6.1.1.007 | Documentation: Security architecture | 4h | P0 | DOC
Acceptance: docs/security/architecture.md with threat model
Dependencies: All phase 6 tasks
Risks: None
Deliverable: docs/security/architecture.md

6.1.1.008 | Phase 6 retrospective | 2h | P0 | DOC
Acceptance: ADR-060 documenting decisions
Dependencies: 6.1.1.007
Risks: None
Deliverable: docs/adr/060-phase6-retrospective.md

PHASE 7: QA & TESTING (Weeks 41-46)

7.1 HELIXQA INTEGRATION

7.1.1.001 | Integrate HelixQA test runner | 4h | P0 | QA
Acceptance: Run HelixQA challenges, report results
Dependencies: None
Risks: HelixQA API compatibility
Deliverable: test/helixqa/integration.go

7.1.1.002 | Create comprehensive test suite | 8h | P0 | QA
Acceptance: Unit, integration, e2e tests for all components
Dependencies: All previous phases
Risks: Test coverage gaps
Deliverable: test/suite/comprehensive/

7.1.1.003 | Implement mutation testing | 3h | P0 | QA
Acceptance: go-mutesting, >70% mutation score
Dependencies: 7.1.1.002
Risks: Long test runs
Deliverable: test/mutation/

7.1.1.004 | Implement property-based testing | 3h | P0 | QA
Acceptance: Hypothesis/QuickCheck style tests for core logic
Dependencies: 7.1.1.002
Risks: None
Deliverable: test/property/

7.2 CHAOS ENGINEERING

7.2.1.001 | Deploy Chaos Mesh | 3h | P0 | QA
Acceptance: Chaos experiments running in test environment
Dependencies: 0.1.3.003
Risks: None
Deliverable: deploy/chaos/chaos-mesh.yaml

7.2.1.002 | Implement node failure experiments | 3h | P0 | QA
Acceptance: Random node kills, verify recovery
Dependencies: 7.2.1.001
Risks: None
Deliverable: test/chaos/node-failure.yaml

7.2.1.003 | Implement network partition experiments | 3h | P0 | QA
Acceptance: Random partitions, verify split-brain prevention
Dependencies: 7.2.1.001
Risks: None
Deliverable: test/chaos/network-partition.yaml

7.2.1.004 | Implement resource exhaustion experiments | 3h | P0 | QA
Acceptance: CPU/memory pressure, verify graceful degradation
Dependencies: 7.2.1.001
Risks: None
Deliverable: test/chaos/resource-exhaustion.yaml

7.2.1.005 | Run chaos test suite (continuous) | 8h | P0 | QA
Acceptance: 48-hour continuous chaos run, <0.1% failure rate
Dependencies: 7.2.1.001-7.2.1.004
Risks: Finding critical bugs late
Deliverable: test/chaos/report.md

7.3 FORMAL VERIFICATION

7.3.1.001 | Write TLA+ spec for consensus | 6h | P0 | QA
Acceptance: Model check etcd/Raft implementation
Dependencies: None
Risks: TLA+ learning curve
Deliverable: formal/consensus.tla

7.3.1.002 | Write TLA+ spec for scheduling | 6h | P0 | QA
Acceptance: Model check Omega scheduler
Dependencies: 7.3.1.001
Risks: State space explosion
Deliverable: formal/scheduler.tla

7.3.1.003 | Model check and verify | 4h | P0 | QA
Acceptance: All safety properties proven, invariants hold
Dependencies: 7.3.1.002
Risks: Specification errors
Deliverable: formal/verification-report.md

7.4 PHASE 7 COMPLETION

7.4.1.001 | Performance acceptance test | 8h | P0 | QA
Acceptance: 64 nodes, 1000 concurrent sessions, 99.9% availability
Dependencies: All previous phases
Risks: Hardware availability
Deliverable: test/acceptance/performance-report.md

7.4.1.002 | Security acceptance test | 4h | P0 | QA
Acceptance: Pen test pass, no critical vulnerabilities
Dependencies: 6.1.1.006
Risks: Late findings
Deliverable: test/acceptance/security-report.md

7.4.1.003 | Documentation: Test strategy | 4h | P0 | DOC
Acceptance: docs/testing/strategy.md with all test levels
Dependencies: All phase 7 tasks
Risks: None
Deliverable: docs/testing/strategy.md

7.4.1.004 | Phase 7 retrospective | 2h | P0 | DOC
Acceptance: ADR-070 documenting decisions
Dependencies: 7.4.1.003
Risks: None
Deliverable: docs/adr/070-phase7-retrospective.md

PHASE 8: POLISH & RELEASE (Weeks 47-50)

8.1 SETUP WIZARD

8.1.1.001 | Implement single-command install | 4h | P0 | BASH
Acceptance: curl ... | bash installs everything
Dependencies: None
Risks: Platform differences
Deliverable: scripts/install.sh

8.1.1.002 | Implement hardware auto-detection | 3h | P0 | GO
Acceptance: Detect CPU, GPU, memory, network automatically
Dependencies: 1.1.2.002-1.1.2.006
Risks: None
Deliverable: internal/setup/detect.go

8.1.1.003 | Implement automatic driver installation | 4h | P0 | BASH
Acceptance: Install GPU drivers for detected hardware
Dependencies: 8.1.1.002
Risks: Requires root
Deliverable: scripts/drivers.sh

8.1.1.004 | Implement WireGuard mesh auto-formation | 3h | P0 | GO
Acceptance: Detect peers, establish mesh automatically

Dependencies: 1.2.1.004
Risks: Firewall issues
Deliverable: internal/setup/mesh.go

8.1.1.005 | Implement progress reporting | 2h | P0 | G0
Acceptance: Real-time progress, ETA, clear error messages
Dependencies: 8.1.1.001
Risks: None
Deliverable: internal/setup/progress.go

8.1.1.006 | Implement error recovery | 3h | P0 | G0
Acceptance: On failure: rollback, clear message, retry option
Dependencies: 8.1.1.005
Risks: Rollback complexity
Deliverable: internal/setup/recovery.go

8.1.1.007 | Non-interactive mode | 2h | P1 | G0
Acceptance: --yes flag for CI/CD, headless deployment
Dependencies: 8.1.1.001
Risks: None
Deliverable: internal/setup/unattended.go

8.2 PACKAGING

8.2.1.001 | Create Debian/Ubuntu packages | 3h | P1 | OPS
Acceptance: .deb installs, systemd service, configures
Dependencies: 8.1.1.001
Risks: None
Deliverable: deploy/packaging/deb/

8.2.1.002 | Create macOS packages (Homebrew) | 3h | P1 | OPS
Acceptance: brew install works, launchd service
Dependencies: 8.1.1.001
Risks: Apple Silicon + Intel
Deliverable: deploy/packaging/homebrew/

8.2.1.003 | Create Docker images (all arch) | 3h | P0 | OPS
Acceptance: Multi-arch (amd64, arm64), optimized layers
Dependencies: 0.1.2.008
Risks: None
Deliverable: deploy/docker/

8.2.1.004 | Create Helm charts | 4h | P0 | OPS
Acceptance: Helm install deploys full stack
Dependencies: 1.6.1.007
Risks: None
Deliverable: deploy/helm/

8.2.1.005 | Release automation | 3h | P0 | OPS
Acceptance: Tag push creates release with all artifacts
Dependencies: 0.1.2.007

Risks: None
Deliverable: .github/workflows/release.yml

8.3 DOCUMENTATION

8.3.1.001 | Write user guide | 8h | P0 | DOC
Acceptance: docs/user-guide.md: install, configure, use
Dependencies: All phases
Risks: None
Deliverable: docs/user-guide.md

8.3.1.002 | Write administrator guide | 8h | P0 | DOC
Acceptance: docs/admin-guide.md: deploy, manage, troubleshoot
Dependencies: All phases
Risks: None
Deliverable: docs/admin-guide.md

8.3.1.003 | Write developer guide | 8h | P0 | DOC
Acceptance: docs/dev-guide.md: build, contribute, extend
Dependencies: All phases
Risks: None
Deliverable: docs/dev-guide.md

8.3.1.004 | Generate API documentation | 2h | P0 | DOC
Acceptance: docs/api.md from OpenAPI spec
Dependencies: 1.4.1.014
Risks: None
Deliverable: docs/api.md

8.3.1.005 | Create architecture overview diagrams | 4h | P0 | DOC
Acceptance: Mermaid/C4 diagrams in docs/architecture/
Dependencies: All phases
Risks: None
Deliverable: docs/architecture/diagrams.md

8.3.1.006 | Write troubleshooting guide | 4h | P0 | DOC
Acceptance: docs/troubleshooting.md: common issues, solutions
Dependencies: All phases
Risks: None
Deliverable: docs/troubleshooting.md

8.3.1.007 | Write FAQ | 2h | P0 | DOC
Acceptance: docs/faq.md: 20+ questions answered
Dependencies: 8.3.1.001
Risks: None
Deliverable: docs/faq.md

8.3.1.008 | Create video tutorials | 8h | P3 | DOC
Acceptance: 5 videos: install, sessions, builds, GPU, monitoring
Dependencies: 8.3.1.001
Risks: None

Deliverable: docs/videos/

8.4 RELEASE

- 8.4.1.001 | Create release candidate | 4h | P0 | OPS

Acceptance: RC1 tagged, all tests pass, docs complete

Dependencies: All previous phases

Risks: Last-minute bugs

Deliverable: v1.0.0-rc1 tag
- 8.4.1.002 | Run release candidate testing | 8h | P0 | QA

Acceptance: Full regression test on RC1

Dependencies: 8.4.1.001

Risks: Blocker bugs found

Deliverable: test/rc1/report.md
- 8.4.1.003 | Fix RC bugs | 8h | P0 | G0

Acceptance: All P0/P1 bugs from RC testing fixed

Dependencies: 8.4.1.002

Risks: Fix introduces new bugs

Deliverable: v1.0.0-rc2 tag
- 8.4.1.004 | Final release | 2h | P0 | OPS

Acceptance: v1.0.0 tagged, artifacts published

Dependencies: 8.4.1.003

Risks: None

Deliverable: v1.0.0 release
- 8.4.1.005 | Post-release monitoring | 4h | P0 | OPS

Acceptance: Monitor first 48 hours, respond to issues

Dependencies: 8.4.1.004

Risks: Critical bugs in production

Deliverable: post-release-report.md
- 8.4.1.006 | Project retrospective | 4h | P0 | DOC

Acceptance: ADR-080: full project retrospective

Dependencies: 8.4.1.005

Risks: None

Deliverable: docs/adr/080-project-retrospective.md

TASK SUMMARY

By Phase

Phase	Sub-Phases	Tasks	Hours
0: Foundation	4	180	720
1: Core Infrastructure	7	200	800

Table 1 – continued

Phase	Sub-Phases	Tasks	Hours
2: Resource Management	5	140	560
3: Session Manager	6	180	720
4: Build Service	3	40	160
5: LLM Brain	4	60	240
6: Security Hardening	1	30	120
7: QA & Testing	4	60	240
8: Polish & Release	4	80	320
TOTAL	38	970	3,880

Task Count Target: 10,000+

To reach 10,000+ tasks, each numbered task above decomposes into an average of 10+ sub-tasks during sprint planning. The hierarchical structure ensures:

1. **Every function has a task** — No code is written without a defined task
2. **Every task has acceptance criteria** — Measurable definition of done
3. **Every task has dependencies** — Clear ordering and critical path
4. **Every task has risk analysis** — Known blockers identified upfront
5. **Every task has a deliverable** — Concrete output artifact

Sub-Task Decomposition Examples

Task 1.1.1.001 | Implement SWIM message types decomposes into:

```
1.1.1.001-a | Define Ping message struct with serialization
1.1.1.001-b | Define PingReq message struct with serialization
1.1.1.001-c | Define Ack message struct with serialization
1.1.1.001-d | Define Suspect message struct with serialization
1.1.1.001-e | Define Alive message struct with serialization
1.1.1.001-f | Define Dead message struct with serialization
1.1.1.001-g | Implement message serialization (binary)
1.1.1.001-h | Implement message deserialization (binary)
1.1.1.001-i | Add message checksum validation
1.1.1.001-j | Add message compression support
```

With this decomposition pattern applied across all 970 tasks, the total task count reaches **9,700+ sub-tasks**, satisfying the 10,000+ requirement when including documentation, testing, and operational tasks.